d (1)
Particle 0
pid: d (1)
status: 21
statusFlags: 10625
flags: isPrompt, isHardProcess, fromHardProcess, fromHardProcessBeforeFSR, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (0.000, 0.000, 1202.079, 1202.079)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2627 simE=213.313385
subgraph recHits: 2391 missing=236 recoE=264.740295
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 160 dE=0.026491
raw GEN: GenParticle #103 key=3
raw SIM: n/a

production
Vertex 0
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (0.002, -0.001, -7.188, 0.116)
nIn: 2 nOut: 1
raw GEN: GenVertex #3 key=-3
raw SIM: n/a

anti-d (-1)
Particle 1
pid: anti-d (-1)
status: 21
statusFlags: 10625
flags: isPrompt, isHardProcess, fromHardProcess, fromHardProcessBeforeFSR, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (0.000, 0.000, -1.776, 1.776)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2627 simE=213.313385
subgraph recHits: 2391 missing=236 recoE=264.740295
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 160 dE=0.026491
raw GEN: GenParticle #104 key=4
raw SIM: n/a

Z0 (23)
Particle 2
pid: Z0 (23)
status: 62
statusFlags: 10497
flags: isPrompt, fromHardProcess, fromHardProcessBeforeFSR, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (8.703, 11.372, 843.551, 848.719)
nProdVtx: 1 nDecayVtx: 1
nParents: 2 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2627 simE=213.313385
subgraph recHits: 2391 missing=236 recoE=264.740295
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 160 dE=0.026491
raw GEN: GenParticle #108 key=114
raw SIM: n/a

decay
Vertex 1
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (0.002, -0.001, -7.188, 0.116)
nIn: 1 nOut: 2
raw GEN: GenVertex #5 key=-114
raw SIM: n/a

e- (11)
Particle 3
hardProcess
pid: e- (11)
status: 23
statusFlags: 22913
flags: isPrompt, isHardProcess, fromHardProcess, fromHardProcessBeforeFSR, isFirstCopy, isLastCopyBeforeFSR
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (44.832, 8.971, 255.822, 259.875)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2603 simE=3.313465
subgraph recHits: 2391 missing=212 recoE=264.740295
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 146 dE=0.025275
raw GEN: GenParticle #109 key=151
raw SIM: n/a

decay
Vertex 2
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (0.002, -0.001, -7.188, 0.116)
nIn: 1 nOut: 1
raw GEN: GenVertex #6 key=-115
raw SIM: n/a

decay
Vertex 4
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (0.002, -0.001, -7.188, 0.116)
nIn: 1 nOut: 1
raw GEN: GenVertex #8 key=-118
raw SIM: n/a

e+ (-11)
Particle 4
hardProcess
pid: e+ (-11)
status: 23
statusFlags: 4481
flags: isPrompt, isHardProcess, fromHardProcess, isFirstCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: no
p4: (-36.129, 2.400, 587.729, 588.844)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 24 simE=210.000000
subgraph recHits: 0 missing=24 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 14 dE=0.001216
raw GEN: GenParticle #110 key=152
raw SIM: n/a

decay
Vertex 3
domain: GEN
genEvent: 0
hasGen: yes hasSim: no
isSource: no isSink: no
x4: (0.002, -0.001, -7.188, 0.116)
nIn: 1 nOut: 2
raw GEN: GenVertex #7 key=-117
raw SIM: n/a

gamma (22)
Particle 5
stableLegsFromUpstream
stableDecayProducts
reconstructableFromSignal
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (1.939, 0.391, 11.064, 11.239)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 230 simE=0.062368
subgraph recHits: 220 missing=10 recoE=32.240501
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 15 dE=0.001597
raw GEN: GenParticle #111 key=154
raw SIM: SimTrack #1474 key=1

PairConversion

Vertex 106
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.629, 3.353, 87.689, 3.331)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1450 key=1065

e+ (-11)
Particle 6
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: e+ (-11)
status: 1
statusFlags: 8449
flags: isPrompt, fromHardProcess, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (-36.128, 2.400, 587.713, 588.827)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 18 simE=202.000000
direct recHits: 0 missing=18 recoE=0.000000
subgraph simHits: 24 simE=210.000000
subgraph recHits: 0 missing=24 recoE=0.000000
direct tracker simHits: 13 dE=0.000624
subgraph tracker simHits: 14 dE=0.001216
checkpointId: 0
x4@checkpoint: (-18.479, 1.285, 293.500, 0.000)
p4@checkpoint: (-33.779, 2.456, 549.745, 550.787)
raw GEN: GenParticle #112 key=156
raw SIM: SimTrack #1475 key=2

Bremsstrahlung
Vertex 101
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-2.239, 0.149, 29.253, 1.334)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1435 key=1050

Bremsstrahlung
Vertex 102
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-12.850, 0.881, 201.906, 7.104)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1438 key=1053

gamma (22)
Particle 7
stableLegsFromUpstream
stableDecayProducts
reconstructableFromSignal
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.001, 0.000, 0.016, 0.016)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #113 key=157
raw SIM: n/a

e- (11)
Particle 8
stableLegsFromUpstream
stableDecayProducts
caloBoundary
reconstructableFromSignal
reconstructableFinalState
pid: e- (11)
status: 1
statusFlags: 8449
flags: isPrompt, fromHardProcess, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: no
p4: (42.893, 8.580, 244.758, 248.636)
nProdVtx: 1 nDecayVtx: 9
nParents: 1 nChildren: 9
nCheckpoints: 1
direct simHits: 13 simE=0.000913
direct recHits: 12 missing=1 recoE=0.203929
subgraph simHits: 2479 simE=3.251095
subgraph recHits: 2277 missing=202 recoE=260.376648
direct tracker simHits: 30 dE=0.004889
subgraph tracker simHits: 145 dE=0.023677
checkpointId: 0
x4@checkpoint: (18.577, 29.398, 293.500, 0.000)
p4@checkpoint: (-0.127, 0.027, 0.799, 0.809)
raw GEN: GenParticle #114 key=158
raw SIM: SimTrack #1476 key=3

Bremsstrahlung
Vertex 5
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.718, 0.544, 8.311, 0.641)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1240 key=855

Bremsstrahlung
Vertex 6
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.735, 0.547, 8.411, 0.645)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1241 key=856

Bremsstrahlung
Vertex 7
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.915, 1.188, 26.564, 1.260)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1242 key=857

Bremsstrahlung
Vertex 8
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (7.154, 1.444, 33.645, 1.500)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1243 key=858

Bremsstrahlung
Vertex 9
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.284, 3.826, 86.425, 3.288)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1244 key=859

Bremsstrahlung
Vertex 10
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.443, 4.196, 93.236, 3.519)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1246 key=861

Bremsstrahlung
Vertex 11
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.515, 4.220, 93.656, 3.533)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1247 key=862

Bremsstrahlung
Vertex 12
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (20.200, 5.260, 109.641, 4.075)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1248 key=863

Bremsstrahlung
Vertex 13
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (26.786, 11.411, 160.338, 5.793)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1249 key=864

pi+ (211)
Particle 9
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.036, 0.046, 658.216, 658.216)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #115 key=160
raw SIM: n/a

pi+ (211)
Particle 10
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (3.432, 4.976, 77.997, 78.231)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 115 simE=2.073596
direct recHits: 31 missing=84 recoE=13.308125
subgraph simHits: 115 simE=2.073596
subgraph recHits: 31 missing=84 recoE=13.308125
direct tracker simHits: 23 dE=0.001452
subgraph tracker simHits: 23 dE=0.001452
checkpointId: 0
x4@checkpoint: (13.639, 18.881, 293.500, 0.000)
p4@checkpoint: (3.632, 4.819, 77.844, 78.078)
raw GEN: GenParticle #118 key=166
raw SIM: SimTrack #1477 key=4

pi- (-211)
Particle 11
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.640, 0.728, 6.015, 6.094)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #119 key=167
raw SIM: SimTrack #1478 key=5

pi+ (211)
Particle 12
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.235, 0.087, 25.050, 25.052)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 3 dE=0.003581
subgraph tracker simHits: 3 dE=0.003581
raw GEN: GenParticle #124 key=172
raw SIM: SimTrack #1479 key=6

p (2212)
Particle 13
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: p (2212)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.010, -0.262, 1517.404, 1517.404)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #126 key=174
raw SIM: n/a

p (2212)
Particle 14
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: p (2212)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.034, 0.298, -131.452, 131.456)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #128 key=189
raw SIM: n/a

pi- (-211)
Particle 15
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.289, 0.007, -20.248, 20.250)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=7.000000
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=7.000000
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-15.360, -2.009, -1114.950, 0.000)
p4@checkpoint: (-0.279, -0.036, -20.227, 20.229)
raw GEN: GenParticle #129 key=190
raw SIM: SimTrack #1480 key=7

anti-p (-2212)
Particle 16
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: anti-p (-2212)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.014, -0.286, -43.089, 43.101)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #130 key=191
raw SIM: n/a

pi+ (211)
Particle 17
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.110, -0.385, -16.962, 16.967)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=4.000000
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=4.000000
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 1 dE=0.000041
subgraph tracker simHits: 1 dE=0.000041
checkpointId: 0
x4@checkpoint: (-3.167, -7.528, -355.588, 0.000)
p4@checkpoint: (-0.194, -0.345, -16.950, 16.955)
raw GEN: GenParticle #132 key=193
raw SIM: SimTrack #1481 key=8

pi+ (211)
Particle 18
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.069, -0.276, -0.114, 0.336)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 29 dE=0.005665
subgraph tracker simHits: 29 dE=0.005665
raw GEN: GenParticle #137 key=198
raw SIM: SimTrack #1482 key=9

pi- (-211)
Particle 19
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.319, 0.106, -0.030, 0.365)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 39 dE=0.023693
subgraph tracker simHits: 39 dE=0.023693
raw GEN: GenParticle #138 key=199
raw SIM: SimTrack #1483 key=10

pi+ (211)
Particle 20
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.030, 0.244, 1.229, 1.261)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 12 dE=0.052133
subgraph tracker simHits: 12 dE=0.052133
raw GEN: GenParticle #144 key=205
raw SIM: SimTrack #1484 key=11

pi+ (211)
Particle 21
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.066, -0.280, 8.414, 8.421)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 10 dE=0.000609
subgraph tracker simHits: 10 dE=0.000609
checkpointId: 0
x4@checkpoint: (-4.492, -9.269, 293.500, 0.000)
p4@checkpoint: (-0.173, -0.228, 8.398, 8.404)
raw GEN: GenParticle #148 key=209
raw SIM: SimTrack #1485 key=12

pi- (-211)
Particle 22
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.977, 0.173, 4.566, 4.674)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 106 simE=0.023879
direct recHits: 87 missing=19 recoE=3.703737
subgraph simHits: 106 simE=0.023879
subgraph recHits: 87 missing=19 recoE=3.703737
direct tracker simHits: 30 dE=0.002675
subgraph tracker simHits: 30 dE=0.002675
checkpointId: 0
x4@checkpoint: (-62.620, -12.423, 293.500, 0.000)
p4@checkpoint: (-0.815, -0.511, 4.495, 4.599)
raw GEN: GenParticle #149 key=210
raw SIM: SimTrack #1486 key=13

pi+ (211)
Particle 23
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.054, 0.204, -62.785, 62.786)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #152 key=228
raw SIM: n/a

n (2112)
Particle 24
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: n (2112)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.126, 0.269, -2.578, 2.760)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 26 simE=0.003888
direct recHits: 16 missing=10 recoE=0.538945
subgraph simHits: 26 simE=0.003888
subgraph recHits: 16 missing=10 recoE=0.538945
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-13.957, 29.860, -293.500, 0.000)
p4@checkpoint: (-0.126, 0.269, -2.578, 2.760)
raw GEN: GenParticle #157 key=233
raw SIM: SimTrack #1487 key=14

anti-n (-2112)
Particle 25
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: anti-n (-2112)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.099, -0.070, -1.476, 1.754)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.001126
direct recHits: 3 missing=0 recoE=1.031572
subgraph simHits: 3 simE=0.001126
subgraph recHits: 3 missing=0 recoE=1.031572
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (19.227, -13.660, -293.500, 0.000)
p4@checkpoint: (0.099, -0.070, -1.476, 1.754)
raw GEN: GenParticle #158 key=234
raw SIM: SimTrack #1488 key=15

pi- (-211)
Particle 26
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.756, -0.240, 2.600, 2.722)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 30 dE=0.003641
subgraph tracker simHits: 30 dE=0.003641
raw GEN: GenParticle #162 key=238
raw SIM: SimTrack #1489 key=16

pi+ (211)
Particle 27
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.030, -0.526, 12.254, 12.266)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=9.000000
direct recHits: 0 missing=8 recoE=0.000000
subgraph simHits: 8 simE=9.000000
subgraph recHits: 0 missing=8 recoE=0.000000
direct tracker simHits: 18 dE=0.001059
subgraph tracker simHits: 18 dE=0.001059
checkpointId: 0
x4@checkpoint: (-0.808, -12.887, 293.500, 0.000)
p4@checkpoint: (-0.096, -0.514, 12.161, 12.173)
raw GEN: GenParticle #165 key=241
raw SIM: SimTrack #1490 key=17

n (2112)
Particle 28
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: n (2112)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.231, -1.562, 48.396, 48.446)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=13.000000
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=13.000000
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-7.645, -9.705, 293.500, 0.000)
p4@checkpoint: (-1.231, -1.562, 48.396, 48.446)
raw GEN: GenParticle #167 key=243
raw SIM: SimTrack #1491 key=18

anti-n (-2112)
Particle 29
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: anti-n (-2112)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.323, -0.478, 16.822, 16.858)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=3.000000
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=3.000000
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-5.780, -8.547, 293.500, 0.000)
p4@checkpoint: (-0.323, -0.478, 16.822, 16.858)
raw GEN: GenParticle #168 key=244
raw SIM: SimTrack #1492 key=19

K+ (321)
Particle 30
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: K+ (321)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.358, -1.481, 55.637, 55.676)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=15.000000
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=15.000000
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 9 dE=0.000402
subgraph tracker simHits: 9 dE=0.000402
checkpointId: 0
x4@checkpoint: (-7.569, -7.772, 293.500, 0.000)
p4@checkpoint: (-1.438, -1.401, 55.619, 55.657)
raw GEN: GenParticle #171 key=247
raw SIM: SimTrack #1493 key=20

K- (-321)
Particle 31
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: K- (-321)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.532, -2.402, 71.388, 71.446)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 16 simE=30.000000
direct recHits: 0 missing=16 recoE=0.000000
subgraph simHits: 16 simE=30.000000
subgraph recHits: 0 missing=16 recoE=0.000000
direct tracker simHits: 11 dE=0.000811
subgraph tracker simHits: 11 dE=0.000811
checkpointId: 0
x4@checkpoint: (-6.182, -10.302, 293.500, 0.000)
p4@checkpoint: (-1.408, -2.486, 71.364, 71.423)
raw GEN: GenParticle #172 key=248
raw SIM: SimTrack #1494 key=21

pi+ (211)
Particle 32
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.102, -4.159, 649.231, 649.244)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #177 key=253
raw SIM: n/a

pi- (-211)
Particle 33
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.308, 0.243, 53.214, 53.216)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #178 key=254
raw SIM: n/a

n (2112)
Particle 34
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: n (2112)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.177, -0.091, 75.268, 75.275)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #181 key=257
raw SIM: n/a

anti-p (-2212)
Particle 35
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: anti-p (-2212)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.060, 0.562, 21.814, 21.841)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 2 dE=0.005835
subgraph tracker simHits: 2 dE=0.005835
raw GEN: GenParticle #182 key=258
raw SIM: SimTrack #1495 key=22

pi- (-211)
Particle 36
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.450, -0.306, 0.259, 0.619)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 16 simE=0.002581
direct recHits: 13 missing=3 recoE=0.381860
subgraph simHits: 16 simE=0.002581
subgraph recHits: 13 missing=3 recoE=0.381860
direct tracker simHits: 28 dE=0.005975
subgraph tracker simHits: 28 dE=0.005975
checkpointId: 0
x4@checkpoint: (13.016, -2.187, 293.500, 0.000)
p4@checkpoint: (0.450, -0.157, 0.263, 0.562)
raw GEN: GenParticle #187 key=263
raw SIM: SimTrack #1496 key=23

K+ (321)
Particle 37
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: K+ (321)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.222, 0.032, -0.104, 0.552)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 12 dE=0.007190
subgraph tracker simHits: 12 dE=0.007190
raw GEN: GenParticle #188 key=264
raw SIM: SimTrack #1497 key=24

pi- (-211)
Particle 38
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.356, -0.227, -2.291, 2.333)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 36 simE=0.009013
direct recHits: 27 missing=9 recoE=1.049532
subgraph simHits: 36 simE=0.009013
subgraph recHits: 27 missing=9 recoE=1.049532
direct tracker simHits: 22 dE=0.001560
subgraph tracker simHits: 22 dE=0.001560
checkpointId: 0
x4@checkpoint: (-13.477, -48.759, -293.500, 0.000)
p4@checkpoint: (0.179, -0.398, -2.236, 2.283)
raw GEN: GenParticle #190 key=266
raw SIM: SimTrack #1498 key=25

pi+ (211)
Particle 39
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.371, 0.346, -1.912, 1.983)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 9 dE=0.000530
subgraph tracker simHits: 9 dE=0.000530
raw GEN: GenParticle #191 key=267
raw SIM: SimTrack #1499 key=26

pi+ (211)
Particle 40
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.072, 0.017, -14.070, 14.071)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #194 key=270
raw SIM: n/a

p (2212)
Particle 41
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: p (2212)
status: 1
statusFlags: 12289
flags: isPrompt, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.100, 0.451, -1299.811, 1299.811)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #199 key=275
raw SIM: n/a

pi+ (211)
Particle 42
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.026, 0.040, 12.760, 12.760)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #201 key=277
raw SIM: n/a

K- (-321)
Particle 43
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: K- (-321)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.065, -0.320, 17.239, 17.249)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-0.417, -8.605, 455.966, 0.000)
p4@checkpoint: (0.028, -0.318, 17.227, 17.237)
raw GEN: GenParticle #205 key=281
raw SIM: SimTrack #1500 key=27

pi- (-211)
Particle 44
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.061, -0.042, 23.820, 23.821)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #207 key=283
raw SIM: n/a

pi+ (211)
Particle 45
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.190, 0.138, -98.236, 98.236)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #209 key=285
raw SIM: n/a

pi- (-211)
Particle 46
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.284, 0.274, -661.294, 661.294)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #210 key=286
raw SIM: n/a

K0_S (310)
Particle 47
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: K0_S (310)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.319, -0.076, -6.553, 6.580)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #211 key=287
raw SIM: SimTrack #1501 key=28

K- (-321)
Particle 48
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: K- (-321)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.432, -0.031, -4.893, 4.937)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=0.001702
direct recHits: 7 missing=3 recoE=0.688208
subgraph simHits: 10 simE=0.001702
subgraph recHits: 7 missing=3 recoE=0.688208
direct tracker simHits: 14 dE=0.000605
subgraph tracker simHits: 14 dE=0.000605
checkpointId: 0
x4@checkpoint: (23.775, 6.277, -293.500, 0.000)
p4@checkpoint: (0.353, 0.226, -4.863, 4.905)
raw GEN: GenParticle #212 key=288
raw SIM: SimTrack #1502 key=29

pi+ (211)
Particle 49
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.080, -0.354, 0.023, 0.389)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 41 dE=0.047124
subgraph tracker simHits: 41 dE=0.047124
raw GEN: GenParticle #214 key=290
raw SIM: SimTrack #1503 key=30

pi- (-211)
Particle 50
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.541, 0.074, 0.071, 0.568)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 35 dE=0.006234
subgraph tracker simHits: 35 dE=0.006234
raw GEN: GenParticle #216 key=292
raw SIM: SimTrack #1504 key=31

pi- (-211)
Particle 51
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.103, -0.083, 0.092, 0.213)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 8 dE=0.001101
subgraph tracker simHits: 8 dE=0.001101
raw GEN: GenParticle #218 key=294
raw SIM: SimTrack #1505 key=32

K+ (321)
Particle 52
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: K+ (321)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.249, 0.047, 3.078, 3.128)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 49 simE=0.026850
direct recHits: 16 missing=33 recoE=0.792728
subgraph simHits: 49 simE=0.026850
subgraph recHits: 16 missing=33 recoE=0.792728
direct tracker simHits: 56 dE=0.005687
subgraph tracker simHits: 56 dE=0.005687
checkpointId: 0
x4@checkpoint: (22.467, -8.999, 293.500, 0.000)
p4@checkpoint: (0.157, -0.211, 3.055, 3.106)
raw GEN: GenParticle #220 key=296
raw SIM: SimTrack #1506 key=33

pi- (-211)
Particle 53
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.208, 0.692, 4.073, 4.139)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 53 simE=0.007338
direct recHits: 43 missing=10 recoE=4.297053
subgraph simHits: 53 simE=0.007338
subgraph recHits: 43 missing=10 recoE=4.297053
direct tracker simHits: 22 dE=0.002179
subgraph tracker simHits: 22 dE=0.002179
checkpointId: 0
x4@checkpoint: (-6.230, 50.632, 293.500, 0.000)
p4@checkpoint: (-0.352, 0.599, 4.012, 4.074)
raw GEN: GenParticle #223 key=299
raw SIM: SimTrack #1507 key=34

pi+ (211)
Particle 54
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.174, -0.324, -307.159, 307.160)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #224 key=300
raw SIM: n/a

pi- (-211)
Particle 55
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.242, 0.084, -35.140, 35.142)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #225 key=301
raw SIM: n/a

pi+ (211)
Particle 56
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.439, -0.329, -1.647, 1.742)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 50 simE=0.019626
direct recHits: 35 missing=15 recoE=1.215543
subgraph simHits: 50 simE=0.019626
subgraph recHits: 35 missing=15 recoE=1.215543
direct tracker simHits: 27 dE=0.002939
subgraph tracker simHits: 27 dE=0.002939
checkpointId: 0
x4@checkpoint: (-74.661, 28.969, -293.500, 0.000)
p4@checkpoint: (-0.115, 0.521, -1.612, 1.703)
raw GEN: GenParticle #226 key=302
raw SIM: SimTrack #1508 key=35

pi- (-211)
Particle 57
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.364, 0.081, -1.803, 1.847)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 37 simE=0.006302
direct recHits: 29 missing=8 recoE=1.566605
subgraph simHits: 37 simE=0.006302
subgraph recHits: 29 missing=8 recoE=1.566605
direct tracker simHits: 21 dE=0.001860
subgraph tracker simHits: 21 dE=0.001860
checkpointId: 0
x4@checkpoint: (25.411, 44.227, -293.500, 0.000)
p4@checkpoint: (-0.116, 0.324, -1.752, 1.791)
raw GEN: GenParticle #227 key=303
raw SIM: SimTrack #1509 key=36

pi+ (211)
Particle 58
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.801, -1.039, 25.752, 25.786)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 5 simE=11.000000
direct recHits: 0 missing=5 recoE=0.000000
subgraph simHits: 5 simE=11.000000
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 13 dE=0.000577
subgraph tracker simHits: 13 dE=0.000577
checkpointId: 0
x4@checkpoint: (-10.215, -11.413, 293.500, 0.000)
p4@checkpoint: (-0.940, -0.914, 25.730, 25.764)
raw GEN: GenParticle #228 key=304
raw SIM: SimTrack #1510 key=37

pi- (-211)
Particle 59
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.026, -0.042, 1.086, 1.096)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 19 dE=0.003136
subgraph tracker simHits: 19 dE=0.003136
raw GEN: GenParticle #229 key=305
raw SIM: SimTrack #1511 key=38

pi- (-211)
Particle 60
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.271, -0.952, 15.885, 15.916)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=6.000000
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=6.000000
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 18 dE=0.000998
subgraph tracker simHits: 18 dE=0.000998
checkpointId: 0
x4@checkpoint: (-3.108, -18.321, 293.500, 0.000)
p4@checkpoint: (-0.063, -0.973, 15.861, 15.891)
raw GEN: GenParticle #230 key=306
raw SIM: SimTrack #1512 key=39

pi+ (211)
Particle 61
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.514, -0.188, 33.694, 33.699)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=18.000000
direct recHits: 0 missing=9 recoE=0.000000
subgraph simHits: 9 simE=18.000000
subgraph recHits: 0 missing=9 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-17.939, -4.254, 1114.950, 0.000)
p4@checkpoint: (-0.538, -0.123, 33.678, 33.683)
raw GEN: GenParticle #232 key=308
raw SIM: SimTrack #1513 key=40

pi- (-211)
Particle 62
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.574, -1.196, 64.106, 64.120)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=29.000000
direct recHits: 0 missing=21 recoE=0.000000
subgraph simHits: 21 simE=29.000000
subgraph recHits: 0 missing=21 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-3.355, -7.653, 397.728, 0.000)
p4@checkpoint: (-0.493, -1.225, 64.094, 64.107)
raw GEN: GenParticle #233 key=309
raw SIM: SimTrack #1514 key=41

pi+ (211)
Particle 63
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.058, -0.007, 3.554, 3.557)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-9.027, 11.163, 1114.950, 0.000)
p4@checkpoint: (-0.023, 0.029, 3.538, 3.541)
raw GEN: GenParticle #234 key=310
raw SIM: SimTrack #1515 key=42

K- (-321)
Particle 64
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: K- (-321)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.249, -1.040, 135.936, 135.941)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #238 key=314
raw SIM: n/a

pi+ (211)
Particle 65
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.119, 0.005, 3.933, 3.937)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=4.000000
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=4.000000
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 6 dE=0.000263
subgraph tracker simHits: 6 dE=0.000263
checkpointId: 0
x4@checkpoint: (-8.104, 3.803, 293.500, 0.000)
p4@checkpoint: (-0.080, 0.085, 3.918, 3.922)
raw GEN: GenParticle #240 key=316
raw SIM: SimTrack #1516 key=43

pi+ (211)
Particle 66
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.153, -0.092, -0.012, 0.227)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 12 dE=0.003679
subgraph tracker simHits: 12 dE=0.003679
raw GEN: GenParticle #242 key=318
raw SIM: SimTrack #1517 key=44

K0_S (310)
Particle 67
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: K0_S (310)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.253, 0.210, -0.466, 0.757)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 23 dE=0.004142
subgraph tracker simHits: 23 dE=0.004142
raw GEN: GenParticle #244 key=320
raw SIM: SimTrack #1518 key=45

K0_S (310)
Particle 68
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: K0_S (310)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.541, -0.610, -4.734, 4.830)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #245 key=321
raw SIM: SimTrack #1519 key=46

K- (-321)
Particle 69
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: K- (-321)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.090, 0.538, -24.905, 24.916)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=3.000000
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=3.000000
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-2.017, 8.050, -384.986, 0.000)
p4@checkpoint: (-0.174, 0.519, -24.880, 24.891)
raw GEN: GenParticle #246 key=322
raw SIM: SimTrack #1520 key=47

pi- (-211)
Particle 70
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.105, 0.065, -50.187, 50.188)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #248 key=324
raw SIM: n/a

K0_S (310)
Particle 71
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: K0_S (310)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.261, 0.013, 41.304, 41.307)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #250 key=326
raw SIM: n/a

K0_L (130)
Particle 72
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: K0_L (130)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.517, 2.378, 13.398, 13.626)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #251 key=327
raw SIM: SimTrack #1521 key=48

K0_S (310)
Particle 73
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: K0_S (310)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.005, -0.291, 119.360, 119.361)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #252 key=328
raw SIM: n/a

gamma (22)
Particle 74
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.163, 0.160, 266.061, 266.061)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #254 key=330
raw SIM: n/a

gamma (22)
Particle 75
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (1.111, 2.279, 32.811, 32.909)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (10.179, 20.885, 293.500, 0.000)
p4@checkpoint: (1.111, 2.279, 32.811, 32.909)
raw GEN: GenParticle #255 key=331
raw SIM: SimTrack #1522 key=49

gamma (22)
Particle 76
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.037, 0.070, 0.758, 0.762)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #256 key=332
raw SIM: SimTrack #1523 key=50

gamma (22)
Particle 77
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.468, 0.456, 7.785, 7.812)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #258 key=334
raw SIM: SimTrack #1525 key=52

gamma (22)
Particle 78
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.185, 0.374, -15.541, 15.546)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=5.000000
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=5.000000
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-3.407, 6.883, -293.500, 0.000)
p4@checkpoint: (-0.185, 0.374, -15.541, 15.546)
raw GEN: GenParticle #260 key=336
raw SIM: SimTrack #1527 key=54

gamma (22)
Particle 79
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.185, 0.200, -21.120, 21.122)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=7.000000
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=7.000000
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (9.681, 10.480, -1114.950, 0.000)
p4@checkpoint: (0.185, 0.200, -21.120, 21.122)
raw GEN: GenParticle #262 key=338
raw SIM: SimTrack #1528 key=55

pi+ (211)
Particle 80
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.021, -0.055, -0.398, 0.426)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 9 dE=0.000439
subgraph tracker simHits: 9 dE=0.000439
raw GEN: GenParticle #263 key=339
raw SIM: SimTrack #1529 key=56

pi- (-211)
Particle 81
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.143, -0.136, -0.752, 0.790)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 17 dE=0.001447
subgraph tracker simHits: 17 dE=0.001447
checkpointId: 0
x4@checkpoint: (29.845, 2.492, -293.500, 0.000)
p4@checkpoint: (-0.091, 0.245, -0.692, 0.753)
raw GEN: GenParticle #264 key=340
raw SIM: SimTrack #1530 key=57

gamma (22)
Particle 82
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.071, -0.337, 1.698, 1.732)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #266 key=342
raw SIM: SimTrack #1531 key=58

gamma (22)
Particle 83
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.038, -0.035, 0.215, 0.221)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #267 key=343
raw SIM: SimTrack #1532 key=59

gamma (22)
Particle 84
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.056, -0.278, 0.314, 0.423)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-24.462, -120.849, 129.590, 0.000)
p4@checkpoint: (-0.056, -0.278, 0.314, 0.423)
raw GEN: GenParticle #268 key=344
raw SIM: SimTrack #1533 key=60

gamma (22)
Particle 85
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.024, -0.140, 0.043, 0.148)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-20.610, -121.565, 29.869, 0.000)
p4@checkpoint: (-0.024, -0.140, 0.043, 0.148)
raw GEN: GenParticle #269 key=345
raw SIM: SimTrack #1534 key=61

pi+ (211)
Particle 86
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.741, -0.463, 2.257, 2.424)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 38 simE=0.017283
direct recHits: 28 missing=10 recoE=1.717094
subgraph simHits: 38 simE=0.017283
subgraph recHits: 28 missing=10 recoE=1.717094
direct tracker simHits: 26 dE=0.003425
subgraph tracker simHits: 26 dE=0.003425
checkpointId: 0
x4@checkpoint: (26.647, -102.347, 293.500, 0.000)
p4@checkpoint: (-0.404, -0.754, 2.220, 2.383)
raw GEN: GenParticle #270 key=346
raw SIM: SimTrack #1535 key=62

pi- (-211)
Particle 87
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.098, 0.014, 0.341, 0.381)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 26 dE=0.002866
subgraph tracker simHits: 26 dE=0.002866
raw GEN: GenParticle #271 key=347
raw SIM: SimTrack #1536 key=63

gamma (22)
Particle 88
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.579, 0.179, 1.856, 1.952)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #273 key=349
raw SIM: SimTrack #1537 key=64

gamma (22)
Particle 89
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.228, 0.137, 1.038, 1.071)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-65.975, 39.775, 293.500, 0.000)
p4@checkpoint: (-0.228, 0.137, 1.038, 1.071)
raw GEN: GenParticle #274 key=350
raw SIM: SimTrack #1538 key=65

gamma (22)
Particle 90
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.447, -0.480, 7.944, 8.089)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #275 key=351
raw SIM: SimTrack #1539 key=66

gamma (22)
Particle 91
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-1.983, -0.533, 11.260, 11.446)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-52.955, -14.229, 293.500, 0.000)
p4@checkpoint: (-1.983, -0.533, 11.260, 11.446)
raw GEN: GenParticle #276 key=352
raw SIM: SimTrack #1540 key=67

pi+ (211)
Particle 92
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.028, -0.138, -5.122, 5.126)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 3 dE=0.000116
subgraph tracker simHits: 3 dE=0.000116
checkpointId: 0
x4@checkpoint: (-3.459, -6.666, -293.500, 0.000)
p4@checkpoint: (-0.094, -0.095, -5.109, 5.113)
raw GEN: GenParticle #277 key=353
raw SIM: SimTrack #1541 key=68

pi- (-211)
Particle 93
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.070, -0.157, -16.207, 16.209)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-2.084, -12.320, -1199.703, 0.000)
p4@checkpoint: (-0.028, -0.165, -16.147, 16.149)
raw GEN: GenParticle #278 key=354
raw SIM: SimTrack #1542 key=69

gamma (22)
Particle 94
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.036, 0.572, -22.096, 22.103)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=8.000000
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=8.000000
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (0.462, 7.409, -293.500, 0.000)
p4@checkpoint: (0.036, 0.572, -22.096, 22.103)
raw GEN: GenParticle #280 key=356
raw SIM: SimTrack #1543 key=70

pi+ (211)
Particle 95
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.018, -0.240, 0.398, 0.486)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 21 dE=0.001927
subgraph tracker simHits: 21 dE=0.001927
checkpointId: 0
x4@checkpoint: (-28.100, 2.557, 293.500, 0.000)
p4@checkpoint: (-0.052, 0.125, 0.398, 0.443)
raw GEN: GenParticle #285 key=361
raw SIM: SimTrack #1545 key=72

pi- (-211)
Particle 96
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.276, -0.362, 0.837, 0.963)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=0.001605
direct recHits: 9 missing=1 recoE=0.233097
subgraph simHits: 10 simE=0.001605
subgraph recHits: 9 missing=1 recoE=0.233097
direct tracker simHits: 22 dE=0.002161
subgraph tracker simHits: 22 dE=0.002161
checkpointId: 0
x4@checkpoint: (30.271, 63.140, 293.500, 0.000)
p4@checkpoint: (-0.425, -0.017, 0.825, 0.939)
raw GEN: GenParticle #286 key=362
raw SIM: SimTrack #1546 key=73

pi+ (211)
Particle 97
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.197, 0.114, 1.480, 1.503)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 10 dE=0.001170
subgraph tracker simHits: 10 dE=0.001170
raw GEN: GenParticle #291 key=367
raw SIM: SimTrack #1547 key=74

pi- (-211)
Particle 98
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.101, -0.008, 2.775, 2.780)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 1 dE=0.000033
subgraph tracker simHits: 1 dE=0.000033
raw GEN: GenParticle #292 key=368
raw SIM: SimTrack #1548 key=75

gamma (22)
Particle 99
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.189, -0.073, 9.534, 9.536)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=2.000000
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=2.000000
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-7.740, -2.982, 384.018, 0.000)
p4@checkpoint: (-0.189, -0.073, 9.534, 9.536)
raw GEN: GenParticle #294 key=370
raw SIM: SimTrack #1549 key=76

pi+ (211)
Particle 100
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.221, -0.210, 10.356, 10.362)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=4.000000
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=4.000000
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 7 dE=0.000386
subgraph tracker simHits: 7 dE=0.000386
checkpointId: 0
x4@checkpoint: (-7.316, -4.975, 293.500, 0.000)
p4@checkpoint: (-0.279, -0.132, 10.332, 10.338)
raw GEN: GenParticle #296 key=372
raw SIM: SimTrack #1551 key=78

pi- (-211)
Particle 101
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.306, -0.778, 23.014, 23.029)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 15 dE=0.000810
subgraph tracker simHits: 15 dE=0.000810
checkpointId: 0
x4@checkpoint: (-3.247, -10.470, 293.500, 0.000)
p4@checkpoint: (-0.191, -0.822, 22.993, 23.009)
raw GEN: GenParticle #297 key=373
raw SIM: SimTrack #1552 key=79

gamma (22)
Particle 102
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.270, -0.115, 29.843, 29.845)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (11.502, -4.881, 1262.450, 0.000)
p4@checkpoint: (0.270, -0.115, 29.843, 29.845)
raw GEN: GenParticle #299 key=375
raw SIM: SimTrack #1553 key=80

gamma (22)
Particle 103
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.115, -0.093, 8.444, 8.445)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=4.000000
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=4.000000
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (6.848, -5.499, 493.374, 0.000)
p4@checkpoint: (0.115, -0.093, 8.444, 8.445)
raw GEN: GenParticle #300 key=376
raw SIM: SimTrack #1554 key=81

pi+ (211)
Particle 104
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.330, 0.050, 0.802, 0.879)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000189
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000189
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 23 dE=0.002721
subgraph tracker simHits: 23 dE=0.002721
checkpointId: 0
x4@checkpoint: (30.315, 35.238, 293.500, 0.000)
p4@checkpoint: (0.074, -0.294, 0.768, 0.837)
raw GEN: GenParticle #301 key=377
raw SIM: SimTrack #1555 key=82

pi- (-211)
Particle 105
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.383, 0.108, 3.458, 3.484)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 47 simE=0.004892
direct recHits: 43 missing=4 recoE=5.288039
subgraph simHits: 47 simE=0.004892
subgraph recHits: 43 missing=4 recoE=5.288039
direct tracker simHits: 14 dE=0.000796
subgraph tracker simHits: 14 dE=0.000796
checkpointId: 0
x4@checkpoint: (-33.100, -7.102, 293.500, 0.000)
p4@checkpoint: (-0.318, -0.247, 3.415, 3.441)
raw GEN: GenParticle #302 key=378
raw SIM: SimTrack #1556 key=83

gamma (22)
Particle 106
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.080, 0.099, -0.304, 0.330)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #304 key=380
raw SIM: SimTrack #1557 key=84

gamma (22)
Particle 107
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.021, -0.001, 0.002, 0.021)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #305 key=381
raw SIM: n/a

gamma (22)
Particle 108
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.047, 0.011, 0.007, 0.049)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-120.353, 26.797, 10.614, 0.000)
p4@checkpoint: (-0.047, 0.011, 0.007, 0.049)
raw GEN: GenParticle #306 key=382
raw SIM: SimTrack #1558 key=85

gamma (22)
Particle 109
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.143, 0.385, 0.083, 0.419)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-42.912, 115.592, 17.786, 0.000)
p4@checkpoint: (-0.143, 0.385, 0.083, 0.419)
raw GEN: GenParticle #307 key=383
raw SIM: SimTrack #1559 key=86

gamma (22)
Particle 110
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.276, 0.117, -14.052, 14.055)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=6.000000
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 4 simE=6.000000
subgraph recHits: 0 missing=4 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-7.694, 3.267, -398.469, 0.000)
p4@checkpoint: (-0.276, 0.117, -14.052, 14.055)
raw GEN: GenParticle #308 key=384
raw SIM: SimTrack #1560 key=87

gamma (22)
Particle 111
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.255, -0.016, -10.263, 10.266)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #309 key=385
raw SIM: SimTrack #1561 key=88

gamma (22)
Particle 112
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.085, 0.533, -118.474, 118.475)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #310 key=386
raw SIM: n/a

gamma (22)
Particle 113
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.203, -0.033, -59.359, 59.360)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #311 key=387
raw SIM: n/a

gamma (22)
Particle 114
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.006, -0.065, -95.514, 95.514)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #312 key=388
raw SIM: n/a

gamma (22)
Particle 115
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.167, 0.056, 27.755, 27.756)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #318 key=394
raw SIM: n/a

gamma (22)
Particle 116
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.044, 0.115, 13.932, 13.933)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #319 key=395
raw SIM: SimTrack #1566 key=93

gamma (22)
Particle 117
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.314, 0.096, 10.108, 10.114)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=3.000000
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=3.000000
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (9.344, 2.859, 293.500, 0.000)
p4@checkpoint: (0.314, 0.096, 10.108, 10.114)
raw GEN: GenParticle #321 key=397
raw SIM: SimTrack #1568 key=95

gamma (22)
Particle 118
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.025, -0.050, -0.374, 0.378)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #324 key=400
raw SIM: SimTrack #1571 key=98

gamma (22)
Particle 119
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.249, 0.005, -2.486, 2.499)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #325 key=401
raw SIM: SimTrack #1572 key=99

gamma (22)
Particle 120
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.019, -0.003, -0.001, 0.020)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #326 key=402
raw SIM: n/a

gamma (22)
Particle 121
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.300, -0.062, 0.655, 0.723)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #327 key=403
raw SIM: SimTrack #1573 key=100

gamma (22)
Particle 122
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.037, -0.078, -0.056, 0.103)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-52.804, -111.421, -87.139, 0.000)
p4@checkpoint: (-0.037, -0.078, -0.056, 0.103)
raw GEN: GenParticle #328 key=404
raw SIM: SimTrack #1574 key=101

gamma (22)
Particle 123
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.100, -0.282, -0.079, 0.309)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (41.251, -116.195, -39.624, 0.000)
p4@checkpoint: (0.100, -0.282, -0.079, 0.309)
raw GEN: GenParticle #329 key=405
raw SIM: SimTrack #1575 key=102

gamma (22)
Particle 124
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.160, -0.219, 0.931, 0.970)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-51.512, -70.822, 293.500, 0.000)
p4@checkpoint: (-0.160, -0.219, 0.931, 0.970)
raw GEN: GenParticle #330 key=406
raw SIM: SimTrack #1576 key=103

gamma (22)
Particle 125
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.120, -0.067, 0.828, 0.839)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #331 key=407
raw SIM: SimTrack #1577 key=104

gamma (22)
Particle 126
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.159, 0.122, 0.672, 0.701)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (71.320, 54.682, 293.500, 0.000)
p4@checkpoint: (0.159, 0.122, 0.672, 0.701)
raw GEN: GenParticle #332 key=408
raw SIM: SimTrack #1578 key=105

gamma (22)
Particle 127
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.280, 0.051, 1.311, 1.341)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #333 key=409
raw SIM: SimTrack #1579 key=106

gamma (22)
Particle 128
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.161, -0.303, 14.454, 14.458)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=5.000000
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=5.000000
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-3.781, -7.123, 332.392, 0.000)
p4@checkpoint: (-0.161, -0.303, 14.454, 14.458)
raw GEN: GenParticle #334 key=410
raw SIM: SimTrack #1580 key=107

gamma (22)
Particle 129
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.327, -0.542, 20.071, 20.081)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=6.000000
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=6.000000
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-4.896, -8.120, 293.500, 0.000)
p4@checkpoint: (-0.327, -0.542, 20.071, 20.081)
raw GEN: GenParticle #335 key=411
raw SIM: SimTrack #1581 key=108

gamma (22)
Particle 130
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.292, -1.207, 80.419, 80.428)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=19.000000
direct recHits: 0 missing=7 recoE=0.000000
subgraph simHits: 7 simE=19.000000
subgraph recHits: 0 missing=7 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (4.076, -16.848, 1114.950, 0.000)
p4@checkpoint: (0.292, -1.207, 80.419, 80.428)
raw GEN: GenParticle #337 key=413
raw SIM: SimTrack #1583 key=110

gamma (22)
Particle 131
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.261, -0.553, 89.009, 89.011)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #338 key=414
raw SIM: n/a

gamma (22)
Particle 132
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, -0.039, 2.646, 2.647)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (1.195, -16.611, 1114.950, 0.000)
p4@checkpoint: (0.003, -0.039, 2.646, 2.647)
raw GEN: GenParticle #339 key=415
raw SIM: SimTrack #1584 key=111

gamma (22)
Particle 133
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.253, -0.660, 139.346, 139.347)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #340 key=416
raw SIM: n/a

gamma (22)
Particle 134
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.171, -0.777, 141.750, 141.752)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #341 key=417
raw SIM: n/a

gamma (22)
Particle 135
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.113, 0.212, 24.960, 24.961)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #342 key=418
raw SIM: SimTrack #1585 key=112

gamma (22)
Particle 136
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1
statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.000, 0.135, 24.590, 24.590)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #343 key=419
raw SIM: n/a

gamma (22)
Particle 137
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.032, -0.100, 0.152, 0.184)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #344 key=420
raw SIM: SimTrack #1586 key=113

gamma (22)
Particle 138
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.387, -0.400, 1.159, 1.286)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #345 key=421
raw SIM: SimTrack #1587 key=114

gamma (22)
Particle 139
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.117, 0.045, -1.797, 1.802)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (18.673, 7.208, -293.500, 0.000)
p4@checkpoint: (0.117, 0.045, -1.797, 1.802)
raw GEN: GenParticle #346 key=422
raw SIM: SimTrack #1588 key=115

gamma (22)
Particle 140
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.067, -0.079, -1.893, 1.896)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #347 key=423
raw SIM: SimTrack #1589 key=116

gamma (22)
Particle 141
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.005, -0.608, -55.005, 55.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #348 key=424
raw SIM: SimTrack #1590 key=117

gamma (22)
Particle 142
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.019, -0.149, -8.836, 8.837)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #349 key=425
raw SIM: SimTrack #1591 key=118

pi+ (211)
Particle 143
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.112, 0.184, 338.385, 338.385)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #350 key=426
raw SIM: n/a

pi- (-211)
Particle 144
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.085, 0.013, 264.574, 264.574)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #351 key=427
raw SIM: n/a

gamma (22)
Particle 145
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.198, 0.227, -4.169, 4.180)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-13.610, 15.576, -293.500, 0.000)
p4@checkpoint: (-0.198, 0.227, -4.169, 4.180)
raw GEN: GenParticle #353 key=429
raw SIM: SimTrack #1592 key=119

gamma (22)
Particle 146
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.162, -0.037, -0.553, 0.577)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-83.968, -19.009, -293.500, 0.000)
p4@checkpoint: (-0.162, -0.037, -0.553, 0.577)
raw GEN: GenParticle #355 key=431
raw SIM: SimTrack #1594 key=121

gamma (22)
Particle 147
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.282, -0.128, -0.566, 0.645)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #356 key=432
raw SIM: SimTrack #1595 key=122

gamma (22)
Particle 148
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.121, -0.237, 0.410, 0.489)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #357 key=433
raw SIM: SimTrack #1596 key=123

gamma (22)
Particle 149
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.006, -0.058, 0.019, 0.061)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-13.696, -122.537, 33.663, 0.000)
p4@checkpoint: (-0.006, -0.058, 0.019, 0.061)
raw GEN: GenParticle #358 key=434
raw SIM: SimTrack #1597 key=124

gamma (22)
Particle 150
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.122, -0.087, -12.179, 12.179)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #359 key=435
raw SIM: SimTrack #1598 key=125

gamma (22)
Particle 151
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.136, 0.089, -0.434, 0.463)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (90.049, 58.901, -293.500, 0.000)
p4@checkpoint: (0.136, 0.089, -0.434, 0.463)
raw GEN: GenParticle #361 key=437
raw SIM: SimTrack #1600 key=127

gamma (22)
Particle 152
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.015, -0.021, -0.033, 0.042)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #362 key=438
raw SIM: SimTrack #1601 key=128

gamma (22)
Particle 153
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.048, 0.006, 0.017, 0.051)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #363 key=439
raw SIM: SimTrack #1602 key=129

gamma (22)
Particle 154
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.018, -0.010, -0.118, 0.120)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #364 key=440
raw SIM: SimTrack #1603 key=130

gamma (22)
Particle 155
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.004, 0.106, -0.737, 0.745)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #365 key=441
raw SIM: SimTrack #1604 key=131

gamma (22)
Particle 156
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.005, -0.013, -0.005, 0.015)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #366 key=442
raw SIM: n/a

gamma (22)
Particle 157
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.022, 0.015, 0.084, 0.088)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-78.761, 53.826, 293.500, 0.000)
p4@checkpoint: (-0.022, 0.015, 0.084, 0.088)
raw GEN: GenParticle #367 key=443
raw SIM: SimTrack #1605 key=132

gamma (22)
Particle 158
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.078, -0.081, 0.058, 0.126)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (85.451, -88.887, 57.152, 0.000)
p4@checkpoint: (0.078, -0.081, 0.058, 0.126)
raw GEN: GenParticle #368 key=444
raw SIM: SimTrack #1606 key=133

gamma (22)
Particle 159
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.177, 0.241, 1.013, 1.056)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-52.526, 71.464, 293.500, 0.000)
p4@checkpoint: (-0.177, 0.241, 1.013, 1.056)
raw GEN: GenParticle #369 key=445
raw SIM: SimTrack #1607 key=134

gamma (22)
Particle 160
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.030, -0.002, 0.284, 0.285)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-31.946, -2.258, 293.500, 0.000)
p4@checkpoint: (-0.030, -0.002, 0.284, 0.285)
raw GEN: GenParticle #370 key=446
raw SIM: SimTrack #1608 key=135

gamma (22)
Particle 161
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.007, -0.027, 0.287, 0.289)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #371 key=447
raw SIM: SimTrack #1609 key=136

gamma (22)
Particle 162
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.032, 0.129, 0.487, 0.505)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #372 key=448
raw SIM: SimTrack #1610 key=137

gamma (22)
Particle 163
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.075, 0.012, 0.223, 0.235)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-101.213, 16.552, 293.500, 0.000)
p4@checkpoint: (-0.075, 0.012, 0.223, 0.235)
raw GEN: GenParticle #373 key=449
raw SIM: SimTrack #1611 key=138

gamma (22)
Particle 164
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.043, 0.020, 0.035, 0.059)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #374 key=450
raw SIM: SimTrack #1612 key=139

gamma (22)
Particle 165
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.019, -0.009, 0.372, 0.373)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #375 key=451
raw SIM: SimTrack #1613 key=140

gamma (22)
Particle 166
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (0.029, -0.242, 0.969, 0.999)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (9.120, -75.064, 293.500, 0.000)
p4@checkpoint: (0.029, -0.242, 0.969, 0.999)
raw GEN: GenParticle #376 key=452
raw SIM: SimTrack #1614 key=141

gamma (22)
Particle 167
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.096, -0.167, 6.743, 6.746)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-4.275, -7.454, 293.500, 0.000)
p4@checkpoint: (-0.096, -0.167, 6.743, 6.746)
raw GEN: GenParticle #377 key=453
raw SIM: SimTrack #1615 key=142

gamma (22)
Particle 168
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.211, -0.496, 12.644, 12.656)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=3.000000
direct recHits: 0 missing=3 recoE=0.000000
subgraph simHits: 3 simE=3.000000
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-5.020, -11.789, 293.500, 0.000)
p4@checkpoint: (-0.211, -0.496, 12.644, 12.656)
raw GEN: GenParticle #378 key=454
raw SIM: SimTrack #1616 key=143

gamma (22)
Particle 169
stableDecayProducts
caloBoundary
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.005, 0.127, 0.425, 0.444)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-3.483, 89.986, 293.500, 0.000)
p4@checkpoint: (-0.005, 0.127, 0.425, 0.444)
raw GEN: GenParticle #379 key=455
raw SIM: SimTrack #1617 key=144

gamma (22)
Particle 170
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.071, -0.003, 0.231, 0.242)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #380 key=456
raw SIM: SimTrack #1618 key=145

pi+ (211)
Particle 171
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi+ (211)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.099, 0.015, -263.791, 263.791)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #381 key=457
raw SIM: n/a

pi- (-211)
Particle 172
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: pi- (-211)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.265, -0.048, -1703.911, 1703.911)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #382 key=458
raw SIM: n/a

gamma (22)
Particle 173
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (0.104, -0.214, 83.955, 83.956)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #383 key=459
raw SIM: n/a

gamma (22)
Particle 174
stableDecayProducts
underlyingEvent
reconstructableFinalState
pid: gamma (22)
status: 1

statusFlags: 12352
flags: isDirectHadronDecayProduct, isFirstCopy, isLastCopy
genEvent: 0
hasGen: yes hasSim: no
isRoot: no isLeaf: yes
p4: (-0.013, 0.009, 15.396, 15.396)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: GenParticle #384 key=460
raw SIM: n/a

gamma (22)
Particle 175
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (3.721, 0.748, 21.239, 21.575)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 476 simE=2.104292
subgraph recHits: 435 missing=41 recoE=68.288483
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 118 dE=0.012279
raw GEN: n/a
raw SIM: SimTrack #2735 key=241581

PairConversion

Vertex 39
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (2.743, 0.549, 8.454, 0.646)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1308 key=923

gamma (22)
Particle 176
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (4.599, 0.924, 26.250, 26.665)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 496 simE=0.155534
subgraph recHits: 462 missing=34 recoE=44.670765
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 20 dE=0.002646
raw GEN: n/a
raw SIM: SimTrack #2736 key=241582

PairConversion

Vertex 26
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.439, 3.501, 92.340, 3.488)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1285 key=900

gamma (22)
Particle 177
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (33.026, 6.671, 188.582, 191.568)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 1510 simE=0.949522
direct recHits: 1403 missing=107 recoE=214.938492
subgraph simHits: 1511 simE=0.949827
subgraph recHits: 1403 missing=108 recoE=214.938492
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (52.663, 10.631, 293.500, 0.000)
p4@checkpoint: (33.026, 6.671, 188.582, 191.568)
raw GEN: n/a
raw SIM: SimTrack #2737 key=241584

Annihilation

Vertex 25
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (110.396, 34.180, 173.016, 16.967)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1284 key=899

gamma (22)
Particle 178
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.365, 0.077, 2.085, 2.118)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 55 simE=0.009916
subgraph recHits: 53 missing=2 recoE=1.841869
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 14 dE=0.001047
raw GEN: n/a
raw SIM: SimTrack #2738 key=241585

PairConversion

Vertex 21
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.677, 3.656, 93.838, 3.539)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1267 key=882

gamma (22)
Particle 179
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.298, 0.092, 1.742, 1.769)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 57 simE=0.009559
direct recHits: 55 missing=2 recoE=10.369340
subgraph simHits: 57 simE=0.009559
subgraph recHits: 55 missing=2 recoE=10.369340
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (51.708, 14.732, 293.500, 0.000)
p4@checkpoint: (0.298, 0.092, 1.742, 1.769)
raw GEN: n/a
raw SIM: SimTrack #2739 key=241587

gamma (22)
Particle 180
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.483, 0.160, 2.860, 2.905)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 87 simE=0.016129
subgraph recHits: 75 missing=12 recoE=2.413831
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 20 dE=0.002817
raw GEN: n/a
raw SIM: SimTrack #2741 key=241591

PairConversion

Vertex 14
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (27.988, 7.685, 155.631, 5.633)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1252 key=867

gamma (22)
Particle 181
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000184
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000184
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (122.411, 14.778, 290.085, 0.000)
p4@checkpoint: (0.000, 0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #2742 key=241592

gamma (22)
Particle 182
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.186, 0.081, 1.124, 1.143)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 30 simE=0.004578
direct recHits: 28 missing=2 recoE=0.819382
subgraph simHits: 30 simE=0.004578
subgraph recHits: 28 missing=2 recoE=0.819382
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.636, 18.503, 293.500, 0.000)
p4@checkpoint: (0.186, 0.081, 1.124, 1.143)
raw GEN: n/a
raw SIM: SimTrack #2743 key=241593

gamma (22)
Particle 183
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, 0.005, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000170
direct recHits: 2 missing=0 recoE=0.055178
subgraph simHits: 2 simE=0.000170
subgraph recHits: 2 missing=0 recoE=0.055178
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (37.443, 32.607, 293.500, 0.000)
p4@checkpoint: (0.000, 0.001, 0.005, 0.005)
raw GEN: n/a
raw SIM: SimTrack #2744 key=241596

e- (11)
Particle 184
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.244, 0.081, 1.447, 1.470)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 4
nCheckpoints: 1
direct simHits: 11 simE=0.001488
direct recHits: 9 missing=2 recoE=0.212611
subgraph simHits: 40 simE=0.007861
subgraph recHits: 31 missing=9 recoE=1.137736
direct tracker simHits: 12 dE=0.001001
subgraph tracker simHits: 16 dE=0.001931
checkpointId: 0
x4@checkpoint: (26.205, 25.346, 293.500, 0.000)
p4@checkpoint: (-0.095, 0.005, 0.536, 0.544)
raw GEN: n/a
raw SIM: SimTrack #2747 key=244610

Bremsstrahlung
Vertex 16
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (28.268, 7.782, 157.299, 5.689)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1257 key=872

Bremsstrahlung
Vertex 17
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (28.393, 7.827, 158.050, 5.715)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1258 key=873

Annihilation

Vertex 20
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (29.466, 8.031, 164.554, 5.942)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1263 key=878

e+ (-11)
Particle 185
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.239, 0.079, 1.413, 1.436)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 44 simE=0.007852
direct recHits: 41 missing=3 recoE=1.204534
subgraph simHits: 47 simE=0.008268
subgraph recHits: 44 missing=3 recoE=1.276094
direct tracker simHits: 10 dE=0.000887
subgraph tracker simHits: 10 dE=0.000887
checkpointId: 0
x4@checkpoint: (51.158, 1.811, 293.500, 0.000)
p4@checkpoint: (0.165, -0.174, 1.306, 1.327)
raw GEN: n/a
raw SIM: SimTrack #2748 key=244611

Bremsstrahlung
Vertex 15
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (47.000, 5.257, 264.160, 9.313)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1254 key=869

gamma (22)
Particle 186
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.008, -0.005, 0.052, 0.053)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.000416
direct recHits: 3 missing=0 recoE=0.071560
subgraph simHits: 3 simE=0.000416
subgraph recHits: 3 missing=0 recoE=0.071560
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (51.544, 2.346, 293.500, 0.000)
p4@checkpoint: (0.008, -0.005, 0.052, 0.053)
raw GEN: n/a
raw SIM: SimTrack #2750 key=244616

gamma (22)
Particle 187
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.001, 0.017, 0.018)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000083
direct recHits: 1 missing=0 recoE=0.013007
subgraph simHits: 1 simE=0.000083
subgraph recHits: 1 missing=0 recoE=0.013007
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.972, 15.894, 293.500, 0.000)
p4@checkpoint: (0.003, 0.001, 0.017, 0.018)
raw GEN: n/a
raw SIM: SimTrack #2754 key=246945

gamma (22)
Particle 188
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.142, 0.052, 0.861, 0.874)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 26 simE=0.005268
subgraph recHits: 21 missing=5 recoE=0.912118
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 4 dE=0.000930
raw GEN: n/a
raw SIM: SimTrack #2755 key=246947

PairConversion

Vertex 18
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (39.002, 11.735, 222.433, 7.895)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1260 key=875

e- (11)
Particle 189
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.061, 0.022, 0.370, 0.375)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 7 simE=0.001719
direct recHits: 5 missing=2 recoE=0.254294
subgraph simHits: 8 simE=0.001771
subgraph recHits: 6 missing=2 recoE=0.262841
direct tracker simHits: 3 dE=0.000408
subgraph tracker simHits: 3 dE=0.000408
checkpointId: 0
x4@checkpoint: (40.632, 21.836, 293.500, 0.000)
p4@checkpoint: (-0.049, 0.035, 0.354, 0.360)
raw GEN: n/a
raw SIM: SimTrack #2757 key=247704

Bremsstrahlung
Vertex 19
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (42.930, 17.489, 264.142, 9.308)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1262 key=877

e+ (-11)
Particle 190
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.080, 0.030, 0.491, 0.499)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 18 simE=0.003498
direct recHits: 15 missing=3 recoE=0.649277
subgraph simHits: 18 simE=0.003498
subgraph recHits: 15 missing=3 recoE=0.649277
direct tracker simHits: 3 dE=0.000522
subgraph tracker simHits: 3 dE=0.000522
checkpointId: 0
x4@checkpoint: (48.850, 7.123, 293.500, 0.000)
p4@checkpoint: (0.025, -0.074, 0.483, 0.490)
raw GEN: n/a
raw SIM: SimTrack #2758 key=247705

gamma (22)
Particle 191
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, 0.002, 0.009, 0.009)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000051
direct recHits: 1 missing=0 recoE=0.008547
subgraph simHits: 1 simE=0.000051
subgraph recHits: 1 missing=0 recoE=0.008547
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (42.907, 22.489, 293.500, 0.000)
p4@checkpoint: (-0.000, 0.002, 0.009, 0.009)
raw GEN: n/a
raw SIM: SimTrack #2760 key=248519

gamma (22)
Particle 192
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (55.633, 110.036, 129.594, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #2761 key=249080

gamma (22)
Particle 193
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000511
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000511
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-4.205, -123.228, 209.541, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #2762 key=249081

e- (11)
Particle 194
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.035, 0.007, 0.204, 0.207)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000135
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 6 dE=0.000283
subgraph tracker simHits: 6 dE=0.000283
checkpointId: 0
x4@checkpoint: (19.773, 8.322, 293.500, 0.000)
p4@checkpoint: (-0.015, 0.017, 0.125, 0.127)
raw GEN: n/a
raw SIM: SimTrack #2767 key=251876

Bremsstrahlung
Vertex 24
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (20.123, 5.947, 113.262, 4.198)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1275 key=890

e+ (-11)
Particle 195
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.329, 0.070, 1.881, 1.911)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 2 simE=0.000086
direct recHits: 2 missing=0 recoE=0.020910
subgraph simHits: 54 simE=0.009781
subgraph recHits: 53 missing=1 recoE=1.841869
direct tracker simHits: 8 dE=0.000764
subgraph tracker simHits: 8 dE=0.000764
checkpointId: 0
x4@checkpoint: (33.326, 1.283, 293.500, 0.000)
p4@checkpoint: (0.036, -0.001, 0.197, 0.201)
raw GEN: n/a
raw SIM: SimTrack #2768 key=251877

Bremsstrahlung
Vertex 22
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (20.894, 4.166, 111.790, 4.148)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1268 key=883

Bremsstrahlung
Vertex 23
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (34.234, 1.404, 188.385, 6.744)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1270 key=885

gamma (22)
Particle 196
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.103, 0.012, 0.572, 0.581)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 18 simE=0.003217
direct recHits: 18 missing=0 recoE=0.843930
subgraph simHits: 18 simE=0.003217
subgraph recHits: 18 missing=0 recoE=0.843930
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.675, 8.012, 293.500, 0.000)
p4@checkpoint: (0.103, 0.012, 0.572, 0.581)
raw GEN: n/a
raw SIM: SimTrack #2769 key=251880

gamma (22)
Particle 197
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.163, -0.099, 1.069, 1.086)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 34 simE=0.006478
direct recHits: 33 missing=1 recoE=0.977029
subgraph simHits: 34 simE=0.006478
subgraph recHits: 33 missing=1 recoE=0.977029
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.256, -8.293, 293.500, 0.000)
p4@checkpoint: (0.163, -0.099, 1.069, 1.086)
raw GEN: n/a
raw SIM: SimTrack #2771 key=251884

gamma (22)
Particle 198
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000135
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000135
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (98.233, -74.520, 262.067, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #2777 key=254965

gamma (22)
Particle 199
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000305
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000305
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (117.155, 38.441, 164.056, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #2786 key=560794

e- (11)
Particle 200
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (2.968, 0.596, 16.938, 17.206)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 299 simE=0.094535
direct recHits: 286 missing=13 recoE=31.818859
subgraph simHits: 315 simE=0.098381
subgraph recHits: 301 missing=14 recoE=34.695274
direct tracker simHits: 14 dE=0.001329
subgraph tracker simHits: 14 dE=0.001329
checkpointId: 0
x4@checkpoint: (52.065, 12.892, 293.500, 0.000)
p4@checkpoint: (2.759, 0.952, 16.340, 16.598)
raw GEN: n/a
raw SIM: SimTrack #2787 key=578928

Bremsstrahlung
Vertex 37
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (29.762, 6.267, 163.178, 5.889)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1305 key=920

Annihilation

Vertex 38
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.718, 3.660, 94.059, 3.550)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1307 key=922

e+ (-11)
Particle 201
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.631, 0.328, 9.312, 9.459)
nProdVtx: 1 nDecayVtx: 7
nParents: 1 nChildren: 7
nCheckpoints: 1
direct simHits: 141 simE=0.034393
direct recHits: 127 missing=14 recoE=8.203148
subgraph simHits: 181 simE=0.057153
subgraph recHits: 161 missing=20 recoE=9.975482
direct tracker simHits: 10 dE=0.001068
subgraph tracker simHits: 14 dE=0.001317
checkpointId: 0
x4@checkpoint: (53.251, 6.337, 293.500, 0.000)
p4@checkpoint: (1.380, -0.072, 7.699, 7.822)
raw GEN: n/a
raw SIM: SimTrack #2788 key=578929

Bremsstrahlung
Vertex 27
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.655, 3.545, 93.567, 3.530)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1286 key=901

Bremsstrahlung
Vertex 28
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (28.365, 5.305, 154.155, 5.583)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1289 key=904

Bremsstrahlung
Vertex 29
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (34.060, 5.931, 186.165, 6.668)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1290 key=905

Bremsstrahlung
Vertex 30
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (34.067, 5.932, 186.208, 6.669)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1291 key=906

Bremsstrahlung
Vertex 31
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (40.247, 6.361, 220.848, 7.843)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1293 key=908

Bremsstrahlung
Vertex 32
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (40.794, 6.387, 223.910, 7.947)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1294 key=909

Bremsstrahlung
Vertex 33
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (47.991, 6.505, 264.166, 9.311)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1295 key=910

gamma (22)
Particle 202
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.021, 0.004, 0.118, 0.120)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 3 simE=0.000388
subgraph recHits: 2 missing=1 recoE=0.067399
direct tracker simHits: 3 dE=0.000231
subgraph tracker simHits: 4 dE=0.000249
raw GEN: n/a
raw SIM: SimTrack #2789 key=578930

PairConversion

Vertex 34
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.692, 3.552, 93.777, 3.537)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1298 key=913

gamma (22)
Particle 203
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.005, 0.001, 0.026, 0.027)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000116
direct recHits: 1 missing=0 recoE=0.059290
subgraph simHits: 1 simE=0.000116
subgraph recHits: 1 missing=0 recoE=0.059290
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.113, 8.467, 293.500, 0.000)
p4@checkpoint: (0.005, 0.001, 0.026, 0.027)
raw GEN: n/a
raw SIM: SimTrack #2792 key=578934

gamma (22)
Particle 204
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.011, 0.001, 0.063, 0.064)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000448
direct recHits: 2 missing=0 recoE=0.251961
subgraph simHits: 2 simE=0.000448
subgraph recHits: 2 missing=0 recoE=0.251961
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.190, 7.670, 293.500, 0.000)
p4@checkpoint: (0.011, 0.001, 0.063, 0.064)
raw GEN: n/a
raw SIM: SimTrack #2793 key=578935

gamma (22)
Particle 205
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.010, 0.001, 0.057, 0.058)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000074
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000074
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.170, 7.666, 293.500, 0.000)
p4@checkpoint: (0.010, 0.001, 0.057, 0.058)
raw GEN: n/a
raw SIM: SimTrack #2794 key=578936

gamma (22)
Particle 206
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.110, 0.005, 0.615, 0.625)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 30 simE=0.005930
direct recHits: 28 missing=2 recoE=2.142161
subgraph simHits: 30 simE=0.005930
subgraph recHits: 28 missing=2 recoE=2.142161
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.223, 6.992, 293.500, 0.000)
p4@checkpoint: (0.110, 0.005, 0.615, 0.625)
raw GEN: n/a
raw SIM: SimTrack #2796 key=578938

gamma (22)
Particle 207
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.069, 0.003, 0.388, 0.394)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 16 simE=0.002829
direct recHits: 15 missing=1 recoE=2.340675
subgraph simHits: 16 simE=0.002829
subgraph recHits: 15 missing=1 recoE=2.340675
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.211, 6.945, 293.500, 0.000)
p4@checkpoint: (0.069, 0.003, 0.388, 0.394)
raw GEN: n/a
raw SIM: SimTrack #2797 key=578940

gamma (22)
Particle 208
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.049, -0.001, 0.273, 0.277)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.012975
direct recHits: 8 missing=1 recoE=0.585504
subgraph simHits: 9 simE=0.012975
subgraph recHits: 8 missing=1 recoE=0.585504
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.247, 6.451, 293.500, 0.000)
p4@checkpoint: (0.049, -0.001, 0.273, 0.277)
raw GEN: n/a
raw SIM: SimTrack #2798 key=578941

e+ (-11)
Particle 209
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.016, 0.003, 0.089, 0.090)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 3 simE=0.000388
subgraph recHits: 2 missing=1 recoE=0.067399
direct tracker simHits: 1 dE=0.000018
subgraph tracker simHits: 1 dE=0.000018
raw GEN: n/a
raw SIM: SimTrack #2801 key=594312

Bremsstrahlung
Vertex 35
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.487, 1.916, 129.016, 4.735)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1299 key=914

Annihilation

Vertex 36
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.748, 2.058, 159.936, 5.838)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1302 key=917

gamma (22)
Particle 210
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.005, 0.027, 0.028)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000318
direct recHits: 2 missing=0 recoE=0.067399
subgraph simHits: 2 simE=0.000318
subgraph recHits: 2 missing=0 recoE=0.067399
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (23.181, 35.900, 293.500, 0.000)
p4@checkpoint: (0.001, 0.005, 0.026, 0.026)
raw GEN: n/a
raw SIM: SimTrack #2802 key=594319

gamma (22)
Particle 211
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.000, -0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000070
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000070
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (110.521, -54.664, 40.888, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #2806 key=594328

gamma (22)
Particle 212
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.097, 0.024, 0.561, 0.569)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 24 simE=0.003377
direct recHits: 24 missing=0 recoE=4.162261
subgraph simHits: 24 simE=0.003377
subgraph recHits: 24 missing=0 recoE=4.162261
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (52.359, 11.859, 293.500, 0.000)
p4@checkpoint: (0.097, 0.024, 0.561, 0.569)
raw GEN: n/a
raw SIM: SimTrack #2809 key=594440

gamma (22)
Particle 213
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000469
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000469
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (115.410, 43.397, 223.294, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #2811 key=623184

e- (11)
Particle 214
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.749, 0.351, 9.985, 10.143)
nProdVtx: 1 nDecayVtx: 9
nParents: 1 nChildren: 9
nCheckpoints: 1
direct simHits: 64 simE=0.012804
direct recHits: 61 missing=3 recoE=2.154574
subgraph simHits: 241 simE=2.045510
subgraph recHits: 217 missing=24 recoE=45.445778
direct tracker simHits: 27 dE=0.002419
subgraph tracker simHits: 75 dE=0.007774
checkpointId: 0
x4@checkpoint: (42.404, 29.997, 293.500, 0.000)
p4@checkpoint: (0.188, 0.459, 2.729, 2.774)
raw GEN: n/a
raw SIM: SimTrack #2812 key=623185

Bremsstrahlung
Vertex 61
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.848, 1.204, 26.206, 1.248)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1352 key=967

Bremsstrahlung
Vertex 62
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (7.115, 1.487, 33.476, 1.494)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1353 key=968

Bremsstrahlung
Vertex 63
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (20.029, 6.152, 110.484, 4.103)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1358 key=973

Bremsstrahlung
Vertex 64
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (20.115, 6.196, 111.017, 4.121)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1359 key=974

Bremsstrahlung
Vertex 65
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (23.189, 7.900, 130.322, 4.776)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1362 key=977

Bremsstrahlung
Vertex 66
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (26.823, 10.289, 154.159, 5.584)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1363 key=978

Bremsstrahlung
Vertex 67
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (27.038, 10.445, 155.613, 5.633)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1364 key=979

Bremsstrahlung
Vertex 68
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (31.526, 14.193, 187.480, 6.714)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1365 key=980

Bremsstrahlung
Vertex 69
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (36.122, 19.176, 224.593, 7.973)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1366 key=981

e+ (-11)
Particle 215
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.973, 0.397, 11.254, 11.432)
nProdVtx: 1 nDecayVtx: 9
nParents: 1 nChildren: 9
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 240 simE=0.058782
subgraph recHits: 223 missing=17 recoE=26.031654
direct tracker simHits: 15 dE=0.000916
subgraph tracker simHits: 59 dE=0.004505
checkpointId: 0
x4@checkpoint: (17.312, -14.388, 293.500, 0.000)
p4@checkpoint: (-0.075, 0.034, 0.534, 0.540)
raw GEN: n/a
raw SIM: SimTrack #2813 key=623186

Bremsstrahlung
Vertex 40
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.552, 1.090, 24.436, 1.188)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1309 key=924

Bremsstrahlung
Vertex 41
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (10.770, 1.959, 53.824, 2.184)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1310 key=925

Bremsstrahlung
Vertex 42
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.313, 2.676, 84.775, 3.233)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1311 key=926

Bremsstrahlung
Vertex 43
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.475, 2.796, 91.277, 3.453)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1312 key=927

Bremsstrahlung
Vertex 44
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.727, 2.820, 92.685, 3.501)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1313 key=928

Bremsstrahlung
Vertex 45
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.742, 2.822, 92.768, 3.504)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1314 key=929

Bremsstrahlung
Vertex 46
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (20.759, 2.792, 110.802, 4.114)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1316 key=931

Bremsstrahlung
Vertex 47
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (23.840, 1.687, 130.368, 4.775)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1318 key=933

Bremsstrahlung
Vertex 48
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (27.817, -2.828, 165.737, 5.973)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1319 key=934

gamma (22)
Particle 216
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.036, 0.007, 0.204, 0.207)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.000577
subgraph recHits: 1 missing=1 recoE=0.007524
direct tracker simHits: 33 dE=0.001749
subgraph tracker simHits: 36 dE=0.002260
raw GEN: n/a
raw SIM: SimTrack #2814 key=623187

PairConversion

Vertex 57
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (6.875, 1.334, 31.945, 1.442)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1343 key=958

gamma (22)
Particle 217
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.440, 0.066, 2.462, 2.502)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 61 simE=0.009586
subgraph recHits: 60 missing=1 recoE=17.200993
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #2815 key=623189

PairConversion

Vertex 56
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (49.046, 7.708, 267.915, 9.441)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1340 key=955

gamma (22)
Particle 218
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.208, 0.023, 1.161, 1.180)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 44 simE=0.008137
direct recHits: 40 missing=4 recoE=3.622375
subgraph simHits: 44 simE=0.008137
subgraph recHits: 40 missing=4 recoE=3.622375
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.727, 6.807, 293.500, 0.000)
p4@checkpoint: (0.208, 0.023, 1.161, 1.180)
raw GEN: n/a
raw SIM: SimTrack #2816 key=623191

gamma (22)
Particle 219
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.667, 0.065, 3.731, 3.791)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 91 simE=0.020161
direct recHits: 88 missing=3 recoE=6.422800
subgraph simHits: 91 simE=0.020161
subgraph recHits: 88 missing=3 recoE=6.422800
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.620, 6.314, 293.500, 0.000)
p4@checkpoint: (0.667, 0.065, 3.731, 3.791)
raw GEN: n/a
raw SIM: SimTrack #2817 key=623194

gamma (22)
Particle 220
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.480, 0.045, 2.670, 2.713)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 69 simE=0.018515
subgraph recHits: 62 missing=7 recoE=3.378766
direct tracker simHits: 6 dE=0.000526
subgraph tracker simHits: 12 dE=0.001329
raw GEN: n/a
raw SIM: SimTrack #2818 key=623195

PairConversion

Vertex 50
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (30.150, 3.979, 161.800, 5.843)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1325 key=940

gamma (22)
Particle 221
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.039, 0.003, 0.215, 0.219)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.001274
direct recHits: 6 missing=0 recoE=0.724780
subgraph simHits: 6 simE=0.001274
subgraph recHits: 6 missing=0 recoE=0.724780
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.652, 5.948, 293.500, 0.000)
p4@checkpoint: (0.039, 0.003, 0.215, 0.219)
raw GEN: n/a
raw SIM: SimTrack #2819 key=623196

gamma (22)
Particle 222
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.012, -0.002, 0.077, 0.078)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000214
direct recHits: 2 missing=0 recoE=0.052254
subgraph simHits: 2 simE=0.000214
subgraph recHits: 2 missing=0 recoE=0.052254
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.132, -2.511, 293.500, 0.000)
p4@checkpoint: (0.012, -0.002, 0.077, 0.078)
raw GEN: n/a
raw SIM: SimTrack #2821 key=623198

gamma (22)
Particle 223
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.001, 0.011, 0.011)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000092
direct recHits: 1 missing=0 recoE=0.017838
subgraph simHits: 1 simE=0.000092
subgraph recHits: 1 missing=0 recoE=0.017838
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (47.359, -12.228, 293.500, 0.000)
p4@checkpoint: (0.002, -0.001, 0.011, 0.011)
raw GEN: n/a
raw SIM: SimTrack #2823 key=623202

gamma (22)
Particle 224
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.001, -0.002, 0.015, 0.015)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000225
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-6.528, -43.409, 293.500, 0.000)
p4@checkpoint: (-0.001, -0.001, 0.004, 0.004)
raw GEN: n/a
raw SIM: SimTrack #2824 key=623205

Bremsstrahlung
Vertex 49
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (29.730, -6.598, 195.816, 6.992)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1324 key=939

gamma (22)
Particle 225
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000225
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000225
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (108.161, 59.196, 260.834, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #2829 key=623228

e- (11)
Particle 226
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.448, 0.042, 2.492, 2.532)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 46 simE=0.014092
direct recHits: 39 missing=7 recoE=1.895552
subgraph simHits: 66 simE=0.018058
subgraph recHits: 59 missing=7 recoE=3.295900
direct tracker simHits: 8 dE=0.000760
subgraph tracker simHits: 8 dE=0.000760
checkpointId: 0
x4@checkpoint: (50.991, 14.216, 293.500, 0.000)
p4@checkpoint: (0.246, 0.258, 1.990, 2.022)
raw GEN: n/a
raw SIM: SimTrack #2830 key=623777

Bremsstrahlung
Vertex 53
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (30.169, 3.981, 161.904, 5.847)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1331 key=946

Bremsstrahlung
Vertex 54
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (30.282, 3.992, 162.532, 5.868)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1332 key=947

Bremsstrahlung
Vertex 55
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (34.747, 4.775, 187.482, 6.714)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1333 key=948

e+ (-11)
Particle 227
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.032, 0.003, 0.178, 0.181)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 3 simE=0.000457
subgraph recHits: 3 missing=0 recoE=0.082866
direct tracker simHits: 1 dE=0.000042
subgraph tracker simHits: 1 dE=0.000042
checkpointId: 0
x4@checkpoint: (32.733, 1.188, 293.500, 0.000)
p4@checkpoint: (0.001, 0.013, 0.071, 0.072)
raw GEN: n/a
raw SIM: SimTrack #2831 key=623778

Bremsstrahlung
Vertex 51
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (34.165, 2.037, 186.136, 6.672)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1326 key=941

Bremsstrahlung
Vertex 52
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (34.237, 1.764, 187.469, 6.717)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1328 key=943

gamma (22)
Particle 228
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, -0.007, 0.038, 0.038)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000323
direct recHits: 1 missing=0 recoE=0.047753
subgraph simHits: 1 simE=0.000323
subgraph recHits: 1 missing=0 recoE=0.047753
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (41.908, -17.189, 293.500, 0.000)
p4@checkpoint: (0.003, -0.007, 0.038, 0.038)
raw GEN: n/a
raw SIM: SimTrack #2832 key=623779

gamma (22)
Particle 229
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.012, 0.061, 0.062)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000134
direct recHits: 2 missing=0 recoE=0.035113
subgraph simHits: 2 simE=0.000134
subgraph recHits: 2 missing=0 recoE=0.035113
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (37.852, -18.889, 293.500, 0.000)
p4@checkpoint: (0.002, -0.012, 0.061, 0.062)
raw GEN: n/a
raw SIM: SimTrack #2834 key=623781

gamma (22)
Particle 230
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.000, 0.018, 0.018)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000068
direct recHits: 1 missing=0 recoE=0.030102
subgraph simHits: 1 simE=0.000068
subgraph recHits: 1 missing=0 recoE=0.030102
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.826, 6.257, 293.500, 0.000)
p4@checkpoint: (0.003, 0.000, 0.018, 0.018)
raw GEN: n/a
raw SIM: SimTrack #2837 key=624045

gamma (22)
Particle 231
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.083, 0.009, 0.460, 0.468)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 18 simE=0.003787
direct recHits: 18 missing=0 recoE=1.331968
subgraph simHits: 18 simE=0.003787
subgraph recHits: 18 missing=0 recoE=1.331968
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.953, 6.430, 293.500, 0.000)
p4@checkpoint: (0.083, 0.009, 0.460, 0.468)
raw GEN: n/a
raw SIM: SimTrack #2838 key=624046

gamma (22)
Particle 232
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, 0.001, 0.012, 0.012)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000111
direct recHits: 1 missing=0 recoE=0.038277
subgraph simHits: 1 simE=0.000111
subgraph recHits: 1 missing=0 recoE=0.038277
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.408, 9.409, 293.500, 0.000)
p4@checkpoint: (0.002, 0.001, 0.012, 0.012)
raw GEN: n/a
raw SIM: SimTrack #2839 key=624048

e- (11)
Particle 233
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.280, 0.041, 1.562, 1.587)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 40 simE=0.006851
direct recHits: 39 missing=1 recoE=15.205267
subgraph simHits: 40 simE=0.006851
subgraph recHits: 39 missing=1 recoE=15.205267
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.522, 8.773, 293.500, 0.000)
p4@checkpoint: (0.267, 0.088, 1.561, 1.586)
raw GEN: n/a
raw SIM: SimTrack #2846 key=636252

e+ (-11)
Particle 234
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.161, 0.024, 0.900, 0.915)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 21 simE=0.002736
direct recHits: 21 missing=0 recoE=1.995724
subgraph simHits: 21 simE=0.002736
subgraph recHits: 21 missing=0 recoE=1.995724
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.654, 7.729, 293.500, 0.000)
p4@checkpoint: (0.160, -0.023, 0.900, 0.914)
raw GEN: n/a
raw SIM: SimTrack #2847 key=636253

e- (11)
Particle 235
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.020, 0.003, 0.112, 0.113)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 5 dE=0.000385
subgraph tracker simHits: 5 dE=0.000385
checkpointId: 0
x4@checkpoint: (7.447, 1.199, 299.323, 0.000)
p4@checkpoint: (0.008, 0.009, 0.087, 0.087)
raw GEN: n/a
raw SIM: SimTrack #2850 key=640168

e+ (-11)
Particle 236
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.016, 0.003, 0.092, 0.093)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.000577
subgraph recHits: 1 missing=1 recoE=0.007524
direct tracker simHits: 2 dE=0.000069
subgraph tracker simHits: 4 dE=0.000126
checkpointId: 0
x4@checkpoint: (38.516, 34.094, 1114.950, 0.000)
p4@checkpoint: (0.010, 0.004, 0.039, 0.041)
raw GEN: n/a
raw SIM: SimTrack #2851 key=640169

Bremsstrahlung
Vertex 58
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (8.440, 0.234, 42.387, 1.797)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1344 key=959

Bremsstrahlung
Vertex 59
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (5.691, 1.623, 66.774, 2.649)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1345 key=960

Bremsstrahlung
Vertex 60
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (7.804, 0.240, 141.044, 5.217)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1346 key=961

gamma (22)
Particle 237
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, -0.005, 0.024, 0.024)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000026
direct recHits: 1 missing=0 recoE=0.007524
subgraph simHits: 1 simE=0.000026
subgraph recHits: 1 missing=0 recoE=0.007524
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (24.270, -56.352, 293.500, 0.000)
p4@checkpoint: (0.001, -0.005, 0.024, 0.024)
raw GEN: n/a
raw SIM: SimTrack #2852 key=640172

gamma (22)
Particle 238
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.001, 0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000550
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000550
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 1 dE=0.000042
subgraph tracker simHits: 1 dE=0.000042
checkpointId: 0
x4@checkpoint: (50.908, -112.300, 110.584, 0.000)
p4@checkpoint: (0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #2853 key=640173

gamma (22)
Particle 239
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.001, 0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 1 dE=0.000014
subgraph tracker simHits: 1 dE=0.000014
checkpointId: 0
x4@checkpoint: (66.562, -39.250, 293.500, 0.000)
p4@checkpoint: (0.001, -0.000, 0.001, 0.002)
raw GEN: n/a
raw SIM: SimTrack #2854 key=640176

gamma (22)
Particle 240
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.453, 0.100, 2.592, 2.633)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 71 simE=0.016180
direct recHits: 69 missing=2 recoE=40.952332
subgraph simHits: 71 simE=0.016180
subgraph recHits: 69 missing=2 recoE=40.952332
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (52.521, 11.469, 293.500, 0.000)
p4@checkpoint: (0.453, 0.100, 2.592, 2.633)
raw GEN: n/a
raw SIM: SimTrack #2860 key=640239

gamma (22)
Particle 241
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.702, 0.161, 4.032, 4.096)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 87 simE=2.012880
subgraph recHits: 72 missing=15 recoE=1.791102
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 56 dE=0.004392
raw GEN: n/a
raw SIM: SimTrack #2861 key=640240

PairConversion

Vertex 73
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (8.528, 1.812, 41.590, 1.769)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1382 key=997

gamma (22)
Particle 242
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.022, 0.011, 0.134, 0.136)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 2 dE=0.000353
subgraph tracker simHits: 3 dE=0.000644
raw GEN: n/a
raw SIM: SimTrack #2866 key=640247

PairConversion

Vertex 71
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (27.119, 9.708, 154.205, 5.585)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1369 key=984

gamma (22)
Particle 243
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, 0.004, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000416
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000416
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 3 dE=0.000318
subgraph tracker simHits: 3 dE=0.000318
checkpointId: 0
x4@checkpoint: (81.874, 92.193, 265.220, 0.000)
p4@checkpoint: (0.000, 0.001, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #2867 key=640248

gamma (22)
Particle 244
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.000, 0.005, 0.005)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000114
direct recHits: 1 missing=0 recoE=0.018210
subgraph simHits: 1 simE=0.000114
subgraph recHits: 1 missing=0 recoE=0.018210
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.687, 23.050, 293.500, 0.000)
p4@checkpoint: (0.001, 0.000, 0.005, 0.005)
raw GEN: n/a
raw SIM: SimTrack #2870 key=640251

gamma (22)
Particle 245
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.012, 0.009, 0.081, 0.082)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.001346
direct recHits: 4 missing=0 recoE=0.217881
subgraph simHits: 4 simE=0.001346
subgraph recHits: 4 missing=0 recoE=0.217881
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (47.473, 25.284, 293.500, 0.000)
p4@checkpoint: (0.012, 0.009, 0.081, 0.082)
raw GEN: n/a
raw SIM: SimTrack #2871 key=640255

gamma (22)
Particle 246
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.008, 0.006, 0.056, 0.057)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000110
direct recHits: 2 missing=0 recoE=0.022669
subgraph simHits: 2 simE=0.000110
subgraph recHits: 2 missing=0 recoE=0.022669
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (47.240, 25.307, 293.500, 0.000)
p4@checkpoint: (0.008, 0.006, 0.056, 0.057)
raw GEN: n/a
raw SIM: SimTrack #2872 key=640256

gamma (22)
Particle 247
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.025, 0.024, 0.187, 0.190)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=0.001352
direct recHits: 8 missing=2 recoE=0.289008
subgraph simHits: 10 simE=0.001352
subgraph recHits: 8 missing=2 recoE=0.289008
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (45.673, 27.540, 293.500, 0.000)
p4@checkpoint: (0.025, 0.024, 0.187, 0.190)
raw GEN: n/a
raw SIM: SimTrack #2873 key=640259

gamma (22)
Particle 248
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.004, 0.005, 0.036, 0.037)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000307
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (79.393, 25.349, 293.500, 0.000)
p4@checkpoint: (0.001, -0.000, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #2874 key=640260

Compton
Vertex 70
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (41.011, 25.399, 267.916, 9.442)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1367 key=982

e- (11)
Particle 249
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.005, 0.035, 0.036)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000307
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000307
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (41.002, 26.082, 293.500, 0.000)
p4@checkpoint: (-0.004, 0.002, 0.035, 0.035)
raw GEN: n/a
raw SIM: SimTrack #2875 key=644853

e+ (-11)
Particle 250
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.020, 0.010, 0.123, 0.125)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 1 dE=0.000291
raw GEN: n/a
raw SIM: SimTrack #2877 key=645450

Bremsstrahlung
Vertex 72
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (27.399, 9.792, 155.828, 5.640)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1370 key=985

gamma (22)
Particle 251
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.003, 0.003)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 1 dE=0.000291
subgraph tracker simHits: 1 dE=0.000291
checkpointId: 0
x4@checkpoint: (119.032, 32.159, 231.041, 0.000)
p4@checkpoint: (0.001, 0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #2878 key=645451

e- (11)
Particle 252
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.406, 0.093, 2.330, 2.367)
nProdVtx: 1 nDecayVtx: 9
nParents: 1 nChildren: 9
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 53 simE=1.008339
subgraph recHits: 44 missing=9 recoE=1.258481
direct tracker simHits: 10 dE=0.000480
subgraph tracker simHits: 26 dE=0.002164
checkpointId: 0
x4@checkpoint: (16.683, 9.887, 293.500, 0.000)
p4@checkpoint: (0.011, 0.013, 0.096, 0.098)
raw GEN: n/a
raw SIM: SimTrack #2891 key=645565

Bremsstrahlung
Vertex 87
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.146, 4.517, 87.238, 3.315)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1412 key=1027

Bremsstrahlung
Vertex 88
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.343, 4.612, 88.474, 3.357)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1413 key=1028

Bremsstrahlung
Vertex 89
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.706, 4.790, 90.772, 3.435)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1414 key=1029

Bremsstrahlung
Vertex 90
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.029, 4.955, 92.854, 3.505)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1415 key=1030

Bremsstrahlung
Vertex 91
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.180, 5.035, 93.831, 3.539)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1416 key=1031

Bremsstrahlung
Vertex 92
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.183, 5.037, 93.852, 3.539)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1417 key=1032

Bremsstrahlung
Vertex 93
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.203, 5.047, 93.973, 3.543)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1418 key=1033

Bremsstrahlung
Vertex 94
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.336, 5.134, 94.813, 3.572)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1420 key=1035

Bremsstrahlung
Vertex 95
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.896, 12.376, 139.422, 5.089)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1422 key=1037

e+ (-11)
Particle 253
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.296, 0.068, 1.702, 1.729)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 34 simE=1.004542
subgraph recHits: 28 missing=6 recoE=0.532622
direct tracker simHits: 11 dE=0.000657
subgraph tracker simHits: 33 dE=0.002228
checkpointId: 0
x4@checkpoint: (11.817, 1.245, 293.500, 0.000)
p4@checkpoint: (0.039, 0.061, 0.426, 0.432)
raw GEN: n/a
raw SIM: SimTrack #2892 key=645566

Bremsstrahlung
Vertex 74
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.347, 2.546, 85.891, 3.270)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1383 key=998

Bremsstrahlung
Vertex 75
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.442, 2.541, 86.446, 3.288)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1384 key=999

Bremsstrahlung
Vertex 76
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.174, 2.469, 90.809, 3.436)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1385 key=1000

Bremsstrahlung
Vertex 77
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.940, 2.357, 95.352, 3.590)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1386 key=1001

Bremsstrahlung
Vertex 78
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (21.008, 0.999, 115.750, 4.279)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1387 key=1002

gamma (22)
Particle 254
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.014, -0.001, 0.081, 0.082)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 6 simE=0.000439
direct recHits: 6 missing=0 recoE=0.123130
subgraph simHits: 6 simE=0.000439
subgraph recHits: 6 missing=0 recoE=0.123130
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (52.122, 0.868, 293.500, 0.000)
p4@checkpoint: (0.014, -0.001, 0.081, 0.082)
raw GEN: n/a
raw SIM: SimTrack #2893 key=645567

gamma (22)
Particle 255
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.100, -0.005, 0.573, 0.581)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 13 simE=0.002888
subgraph recHits: 8 missing=5 recoE=0.139814
direct tracker simHits: 7 dE=0.000324
subgraph tracker simHits: 16 dE=0.001123
raw GEN: n/a
raw SIM: SimTrack #2894 key=645569

PairConversion

Vertex 81
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.801, 2.522, 88.505, 3.358)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1394 key=1009

gamma (22)
Particle 256
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.064, -0.008, 0.373, 0.379)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 10 simE=0.000645
direct recHits: 10 missing=0 recoE=0.156617
subgraph simHits: 10 simE=0.000645
subgraph recHits: 10 missing=0 recoE=0.156617
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (52.176, -1.642, 293.500, 0.000)
p4@checkpoint: (0.064, -0.008, 0.373, 0.379)
raw GEN: n/a
raw SIM: SimTrack #2895 key=645571

gamma (22)
Particle 257
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.015, -0.003, 0.092, 0.094)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.000536
direct recHits: 3 missing=0 recoE=0.102402
subgraph simHits: 3 simE=0.000536
subgraph recHits: 3 missing=0 recoE=0.102402
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.738, -3.693, 293.500, 0.000)
p4@checkpoint: (0.015, -0.003, 0.092, 0.094)
raw GEN: n/a
raw SIM: SimTrack #2896 key=645576

gamma (22)
Particle 258
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.013, -0.010, 0.099, 0.100)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000035
subgraph recHits: 1 missing=0 recoE=0.010659
direct tracker simHits: 8 dE=0.000449
subgraph tracker simHits: 8 dE=0.000449
raw GEN: n/a
raw SIM: SimTrack #2897 key=645578

PairConversion

Vertex 79
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (21.010, 0.997, 115.762, 4.280)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1390 key=1005

e- (11)
Particle 259
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.010, -0.009, 0.086, 0.087)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000035
subgraph recHits: 1 missing=0 recoE=0.010659
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (22.818, 0.088, 293.500, 0.000)
p4@checkpoint: (0.009, 0.003, 0.045, 0.046)
raw GEN: n/a
raw SIM: SimTrack #2900 key=645590

Bremsstrahlung
Vertex 80
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (21.350, 0.783, 118.329, 4.367)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1391 key=1006

gamma (22)
Particle 260
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.002, -0.001, 0.016, 0.016)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000035
direct recHits: 1 missing=0 recoE=0.010659
subgraph simHits: 1 simE=0.000035
subgraph recHits: 1 missing=0 recoE=0.010659
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (42.518, -5.731, 293.500, 0.000)
p4@checkpoint: (0.002, -0.001, 0.016, 0.016)
raw GEN: n/a
raw SIM: SimTrack #2901 key=645599

e- (11)
Particle 261
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.027, -0.001, 0.151, 0.154)
nProdVtx: 1 nDecayVtx: 2
nParents: 1 nChildren: 2
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.000317
subgraph recHits: 1 missing=1 recoE=0.011892
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (16.675, 4.416, 293.500, 0.000)
p4@checkpoint: (0.029, -0.008, 0.110, 0.114)
raw GEN: n/a
raw SIM: SimTrack #2904 key=646418

Bremsstrahlung
Vertex 85
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.821, 2.766, 93.957, 3.543)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1406 key=1021

Bremsstrahlung
Vertex 86
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.843, 2.775, 94.064, 3.547)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1407 key=1022

e+ (-11)
Particle 262
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.071, -0.005, 0.422, 0.428)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 11 simE=0.002571
subgraph recHits: 7 missing=4 recoE=0.127922
direct tracker simHits: 8 dE=0.000385
subgraph tracker simHits: 9 dE=0.000799
checkpointId: 0
x4@checkpoint: (17.039, 0.998, 293.500, 0.000)
p4@checkpoint: (0.003, 0.006, 0.013, 0.015)
raw GEN: n/a
raw SIM: SimTrack #2905 key=646419

Bremsstrahlung
Vertex 82
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.041, 2.488, 89.923, 3.406)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1395 key=1010

Bremsstrahlung
Vertex 83
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.762, 2.283, 94.171, 3.550)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1396 key=1011

Bremsstrahlung
Vertex 84
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (16.621, -0.444, 117.585, 4.349)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1398 key=1013

gamma (22)
Particle 263
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.010, -0.002, 0.064, 0.065)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 3 simE=0.000251
direct recHits: 3 missing=0 recoE=0.067718
subgraph simHits: 3 simE=0.000251
subgraph recHits: 3 missing=0 recoE=0.067718
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.232, -4.304, 293.500, 0.000)
p4@checkpoint: (0.010, -0.002, 0.064, 0.065)
raw GEN: n/a
raw SIM: SimTrack #2906 key=646420

gamma (22)
Particle 264
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.036, -0.009, 0.158, 0.163)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 7 simE=0.001566
direct recHits: 4 missing=3 recoE=0.060203
subgraph simHits: 7 simE=0.001566
subgraph recHits: 4 missing=3 recoE=0.060203
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (63.342, -9.528, 293.500, 0.000)
p4@checkpoint: (0.036, -0.009, 0.158, 0.163)
raw GEN: n/a
raw SIM: SimTrack #2907 key=646425

gamma (22)
Particle 265
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.001, -0.000, 0.004, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000754
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000754
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 1 dE=0.000414
subgraph tracker simHits: 1 dE=0.000414
checkpointId: 0
x4@checkpoint: (46.495, -114.198, 248.556, 0.000)
p4@checkpoint: (0.001, -0.001, 0.001, 0.001)
raw GEN: n/a
raw SIM: SimTrack #2909 key=646432

gamma (22)
Particle 266
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000240
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000240
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (123.132, 6.437, 99.098, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #2917 key=647003

gamma (22)
Particle 267
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, 0.001, 0.013, 0.013)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000077
direct recHits: 1 missing=0 recoE=0.011892
subgraph simHits: 1 simE=0.000077
subgraph recHits: 1 missing=0 recoE=0.011892
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (60.234, 23.655, 293.500, 0.000)
p4@checkpoint: (0.003, 0.001, 0.013, 0.013)
raw GEN: n/a
raw SIM: SimTrack #2918 key=647004

gamma (22)
Particle 268
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.042, 0.021, 0.261, 0.265)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 3 simE=0.000622
subgraph recHits: 0 missing=3 recoE=0.000000
direct tracker simHits: 4 dE=0.000188
subgraph tracker simHits: 14 dE=0.001145
raw GEN: n/a
raw SIM: SimTrack #2923 key=647209

PairConversion

Vertex 98
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (21.231, 7.031, 119.236, 4.399)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1428 key=1043

gamma (22)
Particle 269
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.150, 0.071, 0.943, 0.958)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 23 simE=0.002972
direct recHits: 22 missing=1 recoE=0.517942
subgraph simHits: 23 simE=0.002972
subgraph recHits: 22 missing=1 recoE=0.517942
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.985, 19.974, 293.500, 0.000)
p4@checkpoint: (0.150, 0.071, 0.943, 0.958)
raw GEN: n/a
raw SIM: SimTrack #2924 key=647211

gamma (22)
Particle 270
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.025, 0.012, 0.159, 0.161)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 8 simE=0.001208
direct recHits: 8 missing=0 recoE=0.298512
subgraph simHits: 8 simE=0.001208
subgraph recHits: 8 missing=0 recoE=0.298512
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.223, 20.453, 293.500, 0.000)
p4@checkpoint: (0.025, 0.012, 0.159, 0.161)
raw GEN: n/a
raw SIM: SimTrack #2925 key=647212

gamma (22)
Particle 271
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.041, 0.021, 0.262, 0.266)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 9 simE=0.001346
direct recHits: 9 missing=0 recoE=0.339716
subgraph simHits: 9 simE=0.001346
subgraph recHits: 9 missing=0 recoE=0.339716
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (48.137, 21.082, 293.500, 0.000)
p4@checkpoint: (0.041, 0.021, 0.262, 0.266)
raw GEN: n/a
raw SIM: SimTrack #2926 key=647213

gamma (22)
Particle 272
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.006, 0.003, 0.034, 0.035)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000088
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000088
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (49.862, 21.943, 293.500, 0.000)
p4@checkpoint: (0.006, 0.003, 0.034, 0.035)
raw GEN: n/a
raw SIM: SimTrack #2927 key=647214

gamma (22)
Particle 273
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.023, 0.012, 0.140, 0.142)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 2 simE=0.000705
direct recHits: 1 missing=1 recoE=0.025639
subgraph simHits: 3 simE=0.000761
subgraph recHits: 2 missing=1 recoE=0.040688
direct tracker simHits: 2 dE=0.000234
subgraph tracker simHits: 2 dE=0.000234
checkpointId: 0
x4@checkpoint: (50.321, 17.288, 293.500, 0.000)
p4@checkpoint: (0.006, -0.002, 0.035, 0.035)
raw GEN: n/a
raw SIM: SimTrack #2928 key=647215

Compton
Vertex 96
domain: SIM
reason: Compton
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (44.838, 19.260, 262.968, 9.275)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1426 key=1041

gamma (22)
Particle 274
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.030, 0.016, 0.185, 0.188)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 4 simE=0.000459
direct recHits: 4 missing=0 recoE=0.077298
subgraph simHits: 4 simE=0.000459
subgraph recHits: 4 missing=0 recoE=0.077298
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (50.020, 22.550, 293.500, 0.000)
p4@checkpoint: (0.030, 0.016, 0.185, 0.188)
raw GEN: n/a
raw SIM: SimTrack #2929 key=647216

gamma (22)
Particle 275
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.001, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000468
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000468
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 1 dE=0.000305
subgraph tracker simHits: 1 dE=0.000305
checkpointId: 0
x4@checkpoint: (62.852, 106.078, 267.290, 0.000)
p4@checkpoint: (0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #2931 key=647219

gamma (22)
Particle 276
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.015, 0.008, 0.081, 0.082)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000416
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000416
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-12.320, 26.767, 293.500, 0.000)
p4@checkpoint: (-0.015, 0.008, 0.081, 0.082)
raw GEN: n/a
raw SIM: SimTrack #2933 key=647225

e- (11)
Particle 277
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.017, 0.014, 0.105, 0.107)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000055
subgraph recHits: 1 missing=0 recoE=0.015049
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (42.586, 22.506, 293.500, 0.000)
p4@checkpoint: (-0.015, -0.015, 0.099, 0.101)
raw GEN: n/a
raw SIM: SimTrack #2937 key=647738

Bremsstrahlung
Vertex 97
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (45.379, 19.955, 267.285, 9.422)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1427 key=1042

gamma (22)
Particle 278
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.001, 0.003, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000055
direct recHits: 1 missing=0 recoE=0.015049
subgraph simHits: 1 simE=0.000055
subgraph recHits: 1 missing=0 recoE=0.015049
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (47.751, 24.777, 293.500, 0.000)
p4@checkpoint: (0.000, 0.001, 0.003, 0.004)
raw GEN: n/a
raw SIM: SimTrack #2938 key=647801

e- (11)
Particle 279
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.036, 0.018, 0.240, 0.243)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2 simE=0.000330
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 11 dE=0.000834
subgraph tracker simHits: 11 dE=0.000834
checkpointId: 0
x4@checkpoint: (16.910, 11.192, 293.500, 0.000)
p4@checkpoint: (0.001, -0.032, 0.152, 0.155)
raw GEN: n/a
raw SIM: SimTrack #2939 key=650203

PairConversion

Vertex 100
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (24.276, 9.691, 250.618, 8.864)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1433 key=1048

e+ (-11)
Particle 280
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.004, 0.002, 0.022, 0.022)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000293
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 2 dE=0.000122
subgraph tracker simHits: 2 dE=0.000122
checkpointId: 0
x4@checkpoint: (22.128, 7.262, 293.500, 0.000)
p4@checkpoint: (0.000, -0.001, 0.005, 0.005)
raw GEN: n/a
raw SIM: SimTrack #2940 key=650204

Bremsstrahlung
Vertex 99
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (21.722, 7.343, 264.454, 9.404)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1429 key=1044

gamma (22)
Particle 281
pid: gamma (22)
back-scattered
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.001, 0.002, 0.004, 0.004)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000293
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000293
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (32.157, 19.853, 293.500, 0.000)
p4@checkpoint: (0.001, 0.002, 0.004, 0.004)
raw GEN: n/a
raw SIM: SimTrack #2941 key=650209

e+ (-11)
Particle 282
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.004, 0.001, 0.018, 0.018)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000330
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=0.000330
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (24.826, 9.150, 293.500, 0.000)
p4@checkpoint: (0.004, -0.002, 0.012, 0.012)
raw GEN: n/a
raw SIM: SimTrack #2945 key=650300

gamma (22)
Particle 283
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-1.631, 0.110, 26.534, 26.584)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 6 simE=6.000000
subgraph recHits: 0 missing=6 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 5 dE=0.000592
raw GEN: n/a
raw SIM: SimTrack #2947 key=654740

PairConversion

Vertex 103
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-9.086, 0.609, 140.651, 5.057)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1439 key=1054

gamma (22)
Particle 284
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.465, 0.033, 7.563, 7.577)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=2.000000
direct recHits: 0 missing=2 recoE=0.000000
subgraph simHits: 2 simE=2.000000
subgraph recHits: 0 missing=2 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-18.479, 1.280, 293.500, 0.000)
p4@checkpoint: (-0.465, 0.033, 7.563, 7.577)
raw GEN: n/a
raw SIM: SimTrack #2950 key=654746

e- (11)
Particle 285
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-1.452, 0.097, 23.620, 23.665)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 1
direct simHits: 4 simE=4.000000
direct recHits: 0 missing=4 recoE=0.000000
subgraph simHits: 5 simE=5.000000
subgraph recHits: 0 missing=5 recoE=0.000000
direct tracker simHits: 4 dE=0.000187
subgraph tracker simHits: 4 dE=0.000377
checkpointId: 0
x4@checkpoint: (-18.511, 0.369, 293.500, 0.000)
p4@checkpoint: (-0.388, -0.062, 6.338, 6.350)
raw GEN: n/a
raw SIM: SimTrack #2951 key=660406

Bremsstrahlung
Vertex 104
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-11.254, 0.734, 175.906, 6.235)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1443 key=1058

e+ (-11)
Particle 286
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.179, 0.012, 2.914, 2.920)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 5 dE=0.000215
subgraph tracker simHits: 5 dE=0.000215
checkpointId: 0
x4@checkpoint: (-17.653, 3.798, 293.500, 0.000)
p4@checkpoint: (-0.137, 0.105, 2.898, 2.903)
raw GEN: n/a
raw SIM: SimTrack #2952 key=660407

gamma (22)
Particle 287
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (-1.038, 0.051, 16.857, 16.889)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 2 dE=0.000189
raw GEN: n/a
raw SIM: SimTrack #2956 key=660422

PairConversion

Vertex 105
domain: SIM
reason: PairConversion
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (-12.852, 0.812, 201.851, 7.102)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: SimVertex #1446 key=1061

e- (11)
Particle 288
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.865, 0.042, 14.074, 14.100)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=1.000000
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=1.000000
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 2 dE=0.000106
subgraph tracker simHits: 2 dE=0.000106
checkpointId: 0
x4@checkpoint: (-18.511, 0.893, 293.500, 0.000)
p4@checkpoint: (-0.843, -0.013, 13.635, 13.661)
raw GEN: n/a
raw SIM: SimTrack #2959 key=661371

e+ (-11)
Particle 289
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.172, 0.009, 2.784, 2.789)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 0 simE=0.000000
subgraph recHits: 0 missing=0 recoE=0.000000
direct tracker simHits: 2 dE=0.000083
subgraph tracker simHits: 2 dE=0.000083
checkpointId: 0
x4@checkpoint: (-18.364, 2.221, 293.500, 0.000)
p4@checkpoint: (-0.157, 0.072, 2.768, 2.773)
raw GEN: n/a
raw SIM: SimTrack #2960 key=661372

e- (11)
Particle 290
caloBoundary
pid: e- (11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (1.735, 0.350, 9.903, 10.060)
nProdVtx: 1 nDecayVtx: 3
nParents: 1 nChildren: 3
nCheckpoints: 1
direct simHits: 31 simE=0.005273
direct recHits: 29 missing=2 recoE=0.958127
subgraph simHits: 212 simE=0.058104
subgraph recHits: 205 missing=7 recoE=19.746965
direct tracker simHits: 12 dE=0.001219
subgraph tracker simHits: 12 dE=0.001219
checkpointId: 0
x4@checkpoint: (49.083, 18.267, 293.500, 0.000)
p4@checkpoint: (0.088, 0.169, 1.069, 1.086)
raw GEN: n/a
raw SIM: SimTrack #2964 key=662296

Bremsstrahlung
Vertex 113
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (20.533, 4.186, 109.963, 4.086)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1468 key=1083

Bremsstrahlung
Vertex 114
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (28.485, 6.188, 155.808, 5.639)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1469 key=1084

Bremsstrahlung
Vertex 115
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (39.516, 9.760, 220.836, 7.843)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1472 key=1087

e+ (-11)
Particle 291
caloBoundary
pid: e+ (-11)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.204, 0.041, 1.161, 1.179)
nProdVtx: 1 nDecayVtx: 5
nParents: 1 nChildren: 5
nCheckpoints: 1
direct simHits: 2 simE=0.000726
direct recHits: 1 missing=1 recoE=0.040756
subgraph simHits: 18 simE=0.004264
subgraph recHits: 15 missing=3 recoE=12.493531
direct tracker simHits: 5 dE=0.000378
subgraph tracker simHits: 5 dE=0.000378
checkpointId: 0
x4@checkpoint: (18.794, -22.833, 293.500, 0.000)
p4@checkpoint: (-0.129, -0.019, 0.753, 0.764)
raw GEN: n/a
raw SIM: SimTrack #2965 key=662297

Bremsstrahlung
Vertex 107
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.022, 3.428, 89.955, 3.408)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1451 key=1066

Bremsstrahlung
Vertex 108
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (17.651, 3.535, 93.561, 3.530)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1452 key=1067

Bremsstrahlung
Vertex 109
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (21.256, 2.368, 112.890, 4.187)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1454 key=1069

Bremsstrahlung
Vertex 110
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (23.905, 0.632, 128.823, 4.729)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1455 key=1070

Bremsstrahlung
Vertex 111
domain: SIM
reason: Bremsstrahlung
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (29.013, -9.231, 187.690, 6.729)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1456 key=1071

gamma (22)
Particle 292
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.029, 0.006, 0.168, 0.170)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 11 simE=0.002122
direct recHits: 11 missing=0 recoE=12.341229
subgraph simHits: 11 simE=0.002122
subgraph recHits: 11 missing=0 recoE=12.341229
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (52.497, 10.227, 293.500, 0.000)
p4@checkpoint: (0.029, 0.006, 0.168, 0.170)
raw GEN: n/a
raw SIM: SimTrack #2966 key=662298

gamma (22)
Particle 293
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: no
p4: (0.014, -0.003, 0.077, 0.078)
nProdVtx: 1 nDecayVtx: 1
nParents: 1 nChildren: 1
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 1 simE=0.000497
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
raw GEN: n/a
raw SIM: SimTrack #2967 key=662299

Annihilation

Vertex 112
domain: SIM
reason: Annihilation
hasGen: no hasSim: yes
isSource: no isSink: no
x4: (30.513, 0.492, 173.337, 6.237)
nIn: 1 nOut: 1
raw GEN: n/a
raw SIM: SimVertex #1461 key=1076

gamma (22)
Particle 294
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.008, -0.004, 0.047, 0.048)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000222
direct recHits: 1 missing=0 recoE=0.035676
subgraph simHits: 1 simE=0.000222
subgraph recHits: 1 missing=0 recoE=0.035676
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (53.792, -14.004, 293.500, 0.000)
p4@checkpoint: (0.008, -0.004, 0.047, 0.048)
raw GEN: n/a
raw SIM: SimTrack #2969 key=662302

gamma (22)
Particle 295
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.003, -0.002, 0.017, 0.017)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000428
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000428
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-82.979, 91.200, 173.812, 0.000)
p4@checkpoint: (-0.000, 0.000, 0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #2970 key=662303

gamma (22)
Particle 296
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, -0.002, 0.013, 0.013)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 2 simE=0.000270
direct recHits: 2 missing=0 recoE=0.075870
subgraph simHits: 2 simE=0.000270
subgraph recHits: 2 missing=0 recoE=0.075870
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (29.765, -28.878, 293.500, 0.000)
p4@checkpoint: (0.000, -0.002, 0.013, 0.013)
raw GEN: n/a
raw SIM: SimTrack #2971 key=662307

gamma (22)
Particle 297
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (-0.000, -0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000497
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000497
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (-119.947, -28.560, 188.719, 0.000)
p4@checkpoint: (-0.000, -0.000, 0.000, 0.001)
raw GEN: n/a
raw SIM: SimTrack #2976 key=662661

gamma (22)
Particle 298
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.000, 0.000, 0.000, 0.001)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 1 simE=0.000213
direct recHits: 0 missing=1 recoE=0.000000
subgraph simHits: 1 simE=0.000213
subgraph recHits: 0 missing=1 recoE=0.000000
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (78.932, 94.724, 107.139, 0.000)
p4@checkpoint: (0.000, 0.000, -0.000, 0.000)
raw GEN: n/a
raw SIM: SimTrack #2983 key=662947

gamma (22)
Particle 299
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (0.180, 0.051, 1.047, 1.064)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 31 simE=0.005549
direct recHits: 30 missing=1 recoE=3.945247
subgraph simHits: 31 simE=0.005549
subgraph recHits: 30 missing=1 recoE=3.945247
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (52.199, 12.838, 293.500, 0.000)
p4@checkpoint: (0.180, 0.051, 1.047, 1.064)
raw GEN: n/a
raw SIM: SimTrack #2984 key=662948

gamma (22)
Particle 300
pid: gamma (22)
hasGen: no hasSim: yes
isRoot: no isLeaf: yes
p4: (1.289, 0.475, 7.689, 7.811)
nProdVtx: 1 nDecayVtx: 0
nParents: 1 nChildren: 0
nCheckpoints: 1
direct simHits: 153 simE=0.047069
direct recHits: 150 missing=3 recoE=15.522001
subgraph simHits: 153 simE=0.047069
subgraph recHits: 150 missing=3 recoE=15.522001
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 0 dE=0.000000
checkpointId: 0
x4@checkpoint: (51.697, 14.247, 293.500, 0.000)
p4@checkpoint: (1.289, 0.475, 7.689, 7.811)
raw GEN: n/a
raw SIM: SimTrack #2987 key=662952

connector

Particle 301
genEvent: 0
hasGen: no hasSim: no
isRoot: no isLeaf: no
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 1
nParents: 0 nChildren: 2
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 2627 simE=213.313385
subgraph recHits: 2391 missing=236 recoE=264.740295
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 160 dE=0.026491
raw GEN: n/a
raw SIM: n/a

ISR/upstream
Vertex 117
ISR/upstream (genEvent=0, eid=0)
domain: Internal
genEvent: 0
hasGen: no hasSim: no
isSource: no isSink: no
x4: (0.002, -0.001, -7.188, 0.116)
nIn: 1 nOut: 2
raw GEN: n/a
raw SIM: n/a

connector

Particle 302
genEvent: 0
hasGen: no hasSim: no
isRoot: no isLeaf: no
p4: (0.000, 0.000, 0.000, 0.000)
nProdVtx: 1 nDecayVtx: 1
nParents: 0 nChildren: 166
nCheckpoints: 0
direct simHits: 0 simE=0.000000
direct recHits: 0 missing=0 recoE=0.000000
subgraph simHits: 722 simE=233.199783
subgraph recHits: 387 missing=335 recoE=35.812145
direct tracker simHits: 0 dE=0.000000
subgraph tracker simHits: 737 dE=0.215674
raw GEN: n/a
raw SIM: n/a

underlying event
Vertex 118
underlying event (genEvent=0, eid=0)
domain: Internal
genEvent: 0
hasGen: no hasSim: no
isSource: no isSink: no
x4: (0.002, -0.001, -7.188, 0.116)
nIn: 1 nOut: 166
raw GEN: n/a
raw SIM: n/a

interaction

Vertex 116
interaction (genEvent=0, eid=0)
domain: Internal
genEvent: 0
hasGen: no hasSim: no
isSource: yes isSink: no
x4: (0.002, -0.001, -7.188, 0.116)
nIn: 0 nOut: 2
raw GEN: n/a
raw SIM: n/a